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REVIEW

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A comprehensive study of p53 protein

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Abstract

The protein p53 has been extensively investigated since it was found 43 years ago and has become a "guardian of the genome" that regulates the division of cells by preventing the growth of cells and dividing them, that is, inhibits the development of tumors. Initial proof of protein existence by researchers in the mid-1970s was found by altering and regulating the SV40 big T antigen termed the A protein. Researchers demonstrated how viruses play a role in cancer by employing viruses' ability to create T-antigens complex with viral tumors, which was discovered in 1979 following a viral analysis and cancer analog research. Researchers later in the year 1989 explained that in Murine Friend, a virus-caused erythroleukemia, commonly found that p53 was inactivated to suggest that p53 could be a "tumor suppressor gene." The TP53 gene, encoding p53, is one of human cancer's most frequently altered genes. The protein-regulated biological functions of all p53s include cell cycles, apoptosis, senescence, metabolism of the DNA, angiogenesis, cell differentiation, and immunological response. We tried to unfold the history of the p53 protein, which was discovered long back in 1979, that is, 43 years of research on p53, and how p53's function has been developed through time in this article.

KEYWORDS

angiogenesis, apoptosis, guardian of the genome, p53 protein, SV40 T antigen, tumor suppressor gene

1 | INTRODUCTION

Cancer is the uncontrolled development of cells that can spread throughout the body and invade surrounding tissue. "Cancer" is the second leading cause of death worldwide, according to the WHO. By 2020, it is predicted that there would have been an additional 19.3 million cancer cases worldwide (18.1 million without skin cancer melanoma), as well as possibly 10.0 million fatalities (9.9 million without the usage of skin cancer melanoma). In men, lung, prostate, colorectal, stomach, and liver cancer is more frequent than in women, whereas women are more likely to get breast, colorectal, lung, cervical, and thyroid cancers. With 2.3 million (11.7%) newly reported instances of breast cancer, lung cancer (11.4%), colorectal cancer (CRC) (10.0%), prostate cancer (7.3%), and stomach cancer (5.6%) were the most frequently diagnosed conditions. With an expected 1.8 million deaths from cancer, lung cancer remained the leading cause of mortality (18%), followed by colorectal (9.4%), liver (8.3%), stomach (7.7%), and female breast (6.9%) cancers.[1] According to these latest figures, 50 million people are alive within 5 years following a cancer diagnosis. A normal cell's transformation into a malignant cell is a difficult process.[1] p53 was discovered by a group of investigators working independently at Dundee University (UK), Princeton University, and Sloan-Kettering Memorial Hospital in 1979.[2–4] p53 was an absolute non-negligible member of the protein study before it was accepted as a

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key tumor suppressor protein.[5] P53 has been at the forefront of cancer research since it was discovered in 1979. The tumor protein p53 is a 393 amino acid phosphoprotein p53. On the 17th chromosome, the p53 gene is located (17p13.1). The name originates from the protein molecular mass in the 53 kDa range. The p53 protein (also known as TP53) was found in transformed cells as a protein linked to the SV40 (simian vacuolating virus 40) big T antigen. P53 was initially recognized as a protein target for a viral oncogene when it was found, but genetic and functional evidence gathered 10 years later revealed it to be a tumor suppressor.[2,6] p53 is a major player in tumor suppression and the cellular response to DNA damage, according to an altered-associated protein study published in 1979.[7] Several investigations with the protein and later on the p53 gene have shown it to be an oncogene.[8–10] Tumor suppressor genes are responsible for keeping cancer cells in check. When a gene known as "the guardian of the genome" is inactive, it can contribute to cancer's persistence, growth, and spread.

2 | P53 PROTEIN

There are several terms used to describe the TP53 genes and associated proteins like Tumor protein TP53, antigen NY-CO-13, phosphoprotein TP53, cellular tumor antigen TP53, and transformation-related protein P53. Since p53 has been linked to a variety of cellular activities, it has always had a close relationship with malignant development. The discovery is that the p53 quality is usually abnormal or altered in human cancer (nearly 50%).[3] As a tumor suppressor, p53 is activated to halt advancement in the cell cycle and/or lead to apoptosis when stress disorders the DNA. p53 protein not only acts as a transcription factor for genes encoding proapoptotic effector proteins but also plays a role in transcription-independent cellular signaling that leads to cell death via mitochondrial or cytosolic pathways.[11] The mutant p53 gene is linked to more than half of all human malignancies, expanding the clinical potential for both detection and treatment. As a result, determining the precise amount of p53 protein has become a valuable tool for cancer diagnosis and prognosis. Tumourogenesis is a multistage, multigene process. In recent years, a slew of research has found that p53 missense mutations or allele loss are common in a variety of cancers.[12] p53 is divided into two types: wild type and mutant type. The wild-type p53 DNA-binding domain, for example, limits wild-type p53 to specific genomic areas (at p53 sensitive genes), but mutant p53 is not restricted to specific genomic sequences. In some p53-deficient tumor cell lines, expressing wild-type p53 causes spontaneous cell death.[13] The wild-type (wt-p53) p53 protein is a transcription factor that inhibits angiogenesis and stops cell cycle, senescence, and induces apoptosis through regulating downstream target genes (p21, Bax, PTEN, p48, PAI, and others).[13–15] It is also evident that p53 reduces the occurrence of malignancies by triggering apoptosis in cells activated is not obvious what these implications are for p53 function or oncogenes.[16] Immunohistochemistry (IHC) is unable to detect wild-type p53 protein while it has a short half-life and low cell content, whereas IHC identifies it when it has a high level of cell content and a long half-life.[17,18] In-gel retardation experiments using mutant p53 instead of wild-type p53, mutant DNA probe instead of wild-type sequence, or other monoclonal antibodies instead of anti-p53 antibodies failed to discover mobility-shifted complexes. P53 transactivated reporter genes were created using a heterologous minimal promoter and a 39-bp region from the bax gene promoter that contains the p53-binding site consensus sequences. Gel retardation experiments revealed that only wild-type p53 protein, not mutant p53 protein, was able to bind to oligonucleotides corresponding to this area of the bax promoter.[13] The central role of the p53 proteins is shown in Figure 1.

P53 regulates the survival and demise of aberrant cells by executing a variety of downstream signals, including cellular senescence, cell metabolism, inflammation, autophagy, and other biological processes. It does this by receiving, modifying, and relaying information. "p53 also plays a crucial role in determining cell's response to various cellular stress like DNA damage, nutrient deficiency, and hypoxia by inducing gene transcription, which controls the process of cell cycle and programmed cell death (apoptosis)."[19,20] Double-strand breaks (DSBs) created by the introduction of a restriction endonuclease into cells, as well as UV and X-ray DNA lesions, can all promote p53 accumulation, while heat shock does not.[21] Because heat shock protein is thought to play a role in the stabilization of mutant p53 protein, the increase of p53 following chemical or radiation therapy could have been a simple stress response rather than a specialized response to DNA damage.[22] Low-oxygen areas in solid tumors have a worse prognosis than well-oxygenated tumors, regardless of treatment. Other tumor-microenvironmental variables such as vascularization and angiogenesis may operate synergistically to promote apoptosis in vivo, however in vitro data suggests that hypoxia is sufficient. In p53 tumors, hypoxic regions showed 7.2 times more apoptosis than aerobic regions.[22,23] Hypoxic cancers are less radiocurable from a radiobiological standpoint.[24] A strong link has been discovered between skin cancer and UV radiation exposure. The p53 tumor

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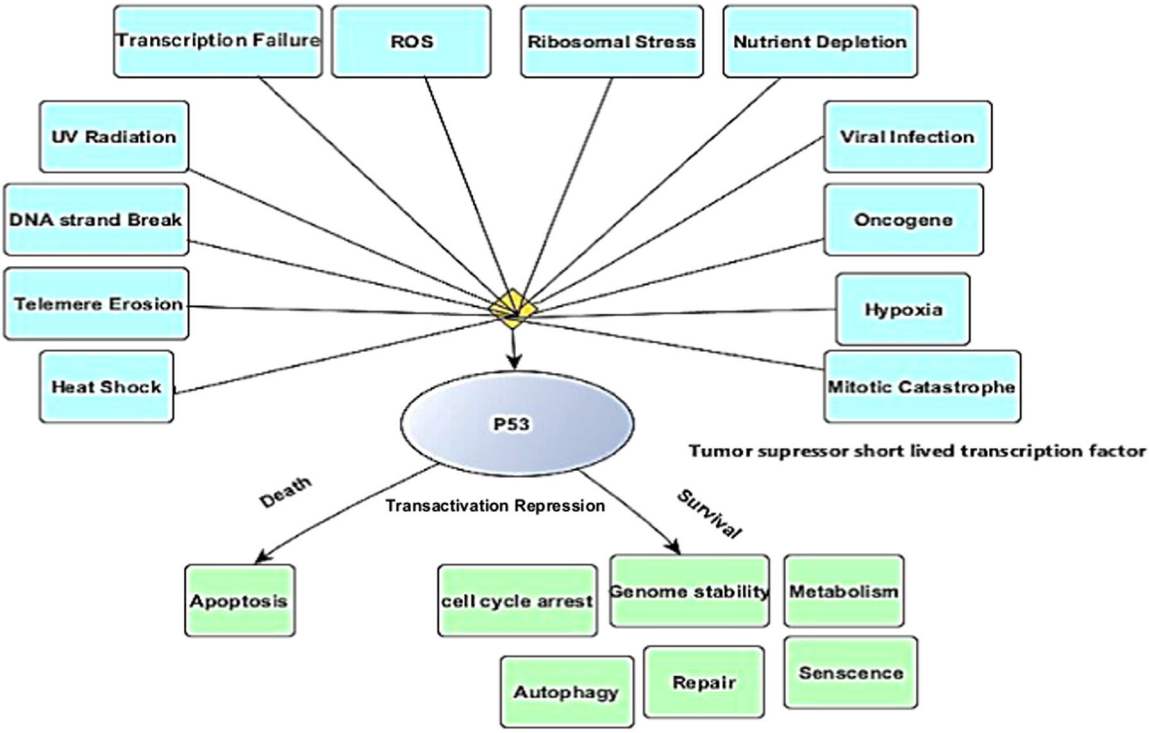

FIGURE 1 Overview of the central role of P53

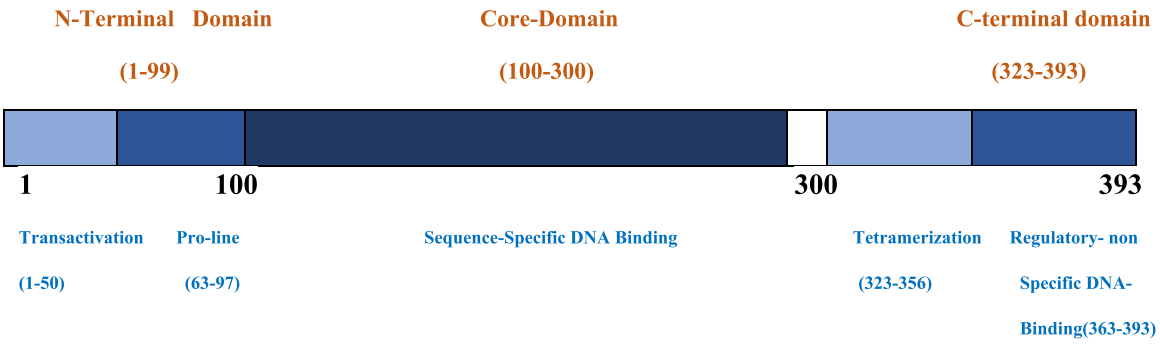

FIGURE 2 Domain organization of the tumor-suppressor protein p53

suppressor gene (also known as TP53) is thought to be an early target in UV radiation-associated skin carcinogenesis because it is frequently mutated in human cancers. compound C-induced apoptosis in the skin depends on the p53 state of the cell. In conclusion, their work demonstrates that C in the lines of the skin cancer cells can produce autophagia and apoptosis.[25] Telomere disruption can lead to cellular senescence and inhibit the growth of tumors, whether it is brought on by too many cell divisions or genotoxic stress. Human fibroblasts replicate again, telomere degradation continues, and the loss of telomere capping function causes widespread chromosomal instability and death, resulting in a cellular state known as crisis.[26]

To fully comprehend the functionality and operation of the p53 protein, we must first comprehend the structure of the protein, the p53 protein sequence, then the function of p53, tumor suppressor function, the apoptosis process, and finally P53 and cell cycle arrest (growth arrest).

3 | P53 PROTEIN STRUCTURE

The p53 gene is approximately 20 kb in length, with 11 exons and 10 introns. It can be located on human chromosome 17's short arm. It has a 2500 bp mRNA. It has 393 amino acids and weighs 53 KDa molecularly. In nearly all somatic cells, the p53 protein is found. p53 is a

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5-domain 393-amino acid protein.[27] The domain organization of the tumor-suppressor of p53 is shown in Figure 2.

A combination of experimental methods, including nuclear magnetic resonance (NMR), electron microscopy, small-angle X-ray scattering, and fluorescence resonance energy transfer (FRET), were used to reveal the structure of the full-length p53 tetramer.[28–32] There are five distinct sections in the p53 domain organization, including transactivation, pro-line, sequence-specific DNA binding, tetramerization, and regulatory regions.

3.1 | Transcriptional activation (transactivation) region (N-terminal [1–42aa])

Different transcription factors, including acetyltransferases and MDM2 (also known as Hdm2 in human beings), needed for amino-terminal transactivity.[33,34] For interaction with transcriptional coactivators and corepressors, the p53 protein's transactivation domain (TAD) is required. TAD1 (residues 1–40) and TAD2 (residues 41–61) are two homologous subdomains found in a variety of transcription-regulating proteins that include conserved Φ-X-X-Φ-Φ sequence motifs (Φ = hydrophobic and X = any amino acid).[35] Phosphorylation inside the TAD generally results in p53 activation and stabilization; however, replacing seven phosphorylated amino acids and theonine residues in TAD1 with alanine leads to a significant loss of p53 transcriptional effects.[36]

3.2 | Proline area (intermediate site [62–97aa])

This is the binding site for p53's negative regulator, the E3 ligase MDM2. Furthermore, at position 72, there is a functional polymorphism that encodes either proline or arginin.[35] The genetic variation is discovered in the proline-rich domain and is responsible for one of the PXXP motifs, suggesting that it has the potential to alter protein-protein interactions.[37] The proline portion of the MDM2 regulator is critical for the stability of p53, and p53 is more susceptible to MDM2 degradation if deleted.[38]

3.3 | DNA binding region (intermediate site [102–292aa])

Amino acid changes in the DNA-binding domain are considerably more likely to cause deactivation than in the NH2- and COOH-terminal regions. The DNA-binding domain's short loops (L1-3 and other loops) that connect the two strands or the two helices were more adaptable to amino acid alteration than the secondary structures (2 α-helices or 11 β-strands). Since the bulk of somatic p53 cancer mutations appears in the DNA-binding domain, the tetramerization area of these mutant proteins is preserved. The formation of mixed tetramers between wild-type and mutant p53 with decreased activity is thought to be the molecular basis for the so-called dominant-negative effect of mutant p53 in heterozygous cells.[39–42] The majority of the protein core consists of the domain of DNA binding required to bind the DNA sequence (the consent sequence includes two copies of 5′-PuPuC(A/T)-(T/a) GPyPy-3′ 10-bp motif, divided by 0–13 bp). Forty percent to fifty percent of tumor cell mutations occur in the p53 protein's intermediate region (102-292aa), which contains four conserved regions.[43]

3.4 | Tetramerization region (323–356aa)

The tetramerization domain of p53 (residues 323–356aa in human p53) controls oligomerization. X-ray crystallography and NMR were used to solve the structure of this domain. A steep turn, made possible by a conserved glycine residue (Gly334), connects these two structural pieces. A tetramer, or dimer of primary dimers, is formed by four chains.[39–42,44,45] Certain residues of the α-helix and β-strands of the tetramerization domain were susceptible to inactivation by amino acid substitutions. "Phe-328, Leu-330, Ile-332, and Arg-333 in the Phe-328, Leu-330, Ile-332, and Arg-333" in the β-strand, "Gly-334 in a turn, and Arg-337, Phe-338, Phe-341,

TABLE 1 P53 protein for different species

Species (sequence)	Human P53	Mouse P53	Rat P53
Length	393	387	391
Mass (Da)	43 653	43 155	43 451
Sequence	Human p53 protein sequence	Mouse p53 protein sequence	Rat p53 protein sequence

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TABLE 2 Essential amino acid in p53 protein (20 amino acid types)

Sr. no	Amino acid	Abbreviation	Single letter abbreviation	Frequency counter from sequence
1	Alanine	Ala	A	24
2	Arginine	Arg	R	26
3	Asparagine	Asn	N	14
4	Aspartic acid	Asp	D	20
5	Cysteine	Cys	C	10
6	Glutamine	Gln	Q	15
7	Glutamic acid	Glu	E	30
8	Glycine	Gly	G	23
9	Histidine	His	H	12
10	Isoleucine	Iso	I	8
11	Leucine	Leu	L	32
12	Lysine	Lys	K	20
13	Methionine	Met	M	12
14	Phenylalanine	Phe	F	11
15	Proline	Pro	P	45
16	Serine	Ser	S	38
17	Threonine	Thr	T	22
18	Tryptophan	Try	W	4
19	Tyrosine	Tyr	Y	9
20	Valine	Val	V	18
Total				393

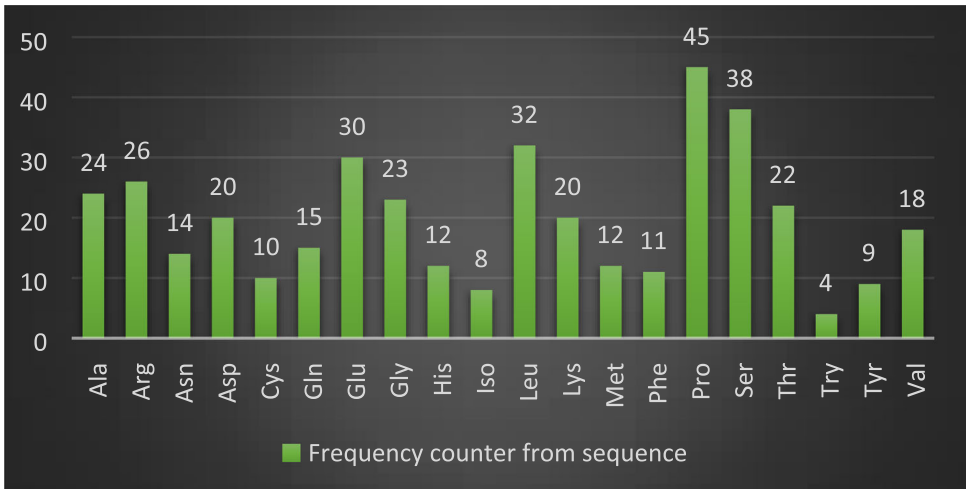

FIGURE 3 Graph for Frequency counter from the amino acid sequence (393 -length)

Phe-342, Leu-344, Asn-345, Ala-347, Leu-348, Leu-350, and Lys-351" in the α-helix are among the 15 residues. Since they were inactivated by at least three different changes, "Leu-330, Ile-332, Gly-334, Arg-337, Phe-341, Leu-344, and Ala-347" were particularly vulnerable to functional inactivation. The hydrophobic residues (Phe, Leu, Ile, or Ala) that make up the hydrophobic core in the domain tend to be inactivated when they are

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replaced with non-hydrophobic residues or structure-destructive proline.[46]

3.5 | Regulatory nonspecific DNA binding region (363–393aa)

The p53 C-terminal domain (CTD) is an intrinsically disordered region that adopts a variety of conformations when it interacts with different binding partners. It has been demonstrated that the basic C-terminus of p53 induces cell death and functions as a negative regulatory domain.[47,48] The N-terminal region of p53 participates in the recruitment of transcriptional proteins. Nonspecific DNA binding activity in the p53 CTD may be crucial for the central core domain's sequence-specific DNA binding.

4 | P53 PROTEIN SEQUENCE

The majority of cancer mutations affect the TP53 gene on chromosome 17p13.1, which produces the p53 protein and is responsible for over half of all cancer changes.[49] The protein's molecular mass of 53 kDa gives it the name p53. The gene is 19 143 base pairs long, starting at base pair 7 571 719 and ending at base pair 7 590 862. Tables 1 and 2 show the description of p53 protein for different species and the essential amino acids in p53 protein types (20 amino acid types) respectively. Later, a 393 amino acid sequence(length) for the protein p53 for *Homo sapiens* was provided. The essential amino acids in p53 protein types are then graphed in Figure 3.

4.1 | Sequence for the P53 protein (*Homo sapiens*)

>sp|P04637|P53_HUMAN Cellular tumor antigen p53 OS = *Homo sapiens* OX = 9606 GN = TP53 PE = 1 SV = 4.

MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLP
SQAMDDLMLSPDDIEQWFTEDPGPDEAPRMPEAAPP-
VAPAPAAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGFR
LGFLHSGTAKSVTCTYSPALNKMFCQLAKTCPVQLW
VDSTPPPGTRVRAMAIYKQSQHMTEVVRRCPHHERC
SDSDGLAPPQHLIRVEGNLRVEYLDDRNTFRHSVVVP
YEPPEVGSDCTTIHYNYMCNSSCMGGMNRRPILTIITL
EDSSGNLLGRNSFEVRVCACPGRDRRTEEENLRKKGE
PHHELPPGSTKRALPNNTSSSPQPKKKPLDGEYFTLQI
RGRERFEMFRELNEALELKDAQAGKEPGGSRAHSS
HLKSKKGQSTSRHKKLMFKTEGPDSD.

5 | THE FUNCTIONS OF P53 PROTEIN

p53 protein is required for maintaining genome integrity and inhibiting uncontrolled cell growth.[43,47] The wild-type P53, a crucial regulatory protein, is activated by heat shock, hypoxia, and oncogenic overexpression and triggers several biological responses, both on a single cell and a whole organism level.[16,47,49,50] Increased overall quantities of p53 proteins, along with qualitative protein alterations induced by major conformational changes, activate the genes p53-targeted.[51] When DNA damage results in DSBs, the ATM protein kinase (ataxia telangiectasia mutated) is activated, which then activates Chk2 kinase.[52] ATM and Chk2 then lead to the activation of p53 at various sites in p53-dependent cells, leading to growth inhibition of apoptosis.[53,54] Furthermore, DNA damage can cause replication obstruction, resulting in the activation of the ATR kinase. As a result, phosphorylate and activated p53 both activated ATR and then activated Chk1.[55,56] Cell-cycle phases, cell survival, death, and senescence are all triggered by wild-type p53 activating a wide range of genes that act as death receptors of signaling pathways.[19,57] Numerous roles of p53 are related to the ability to function as a sequence-specific transcription factor that affects multiple cellular processes, including the crucial role in tumor elimination.[58] p53 accumulates in the nucleus as a reaction to different types of stress, links to certain locations in p53-responding gene regulation areas, and then raises its production considerably.[43,59] In numerous experimental contexts, a variety of techniques have been employed to find the p53 goals. Depending on the kind of cell and the degree of p53 damage, the downstream p53 targets are activated differently, and other factors are presently being studied.[60] Hundreds of functionally p53-dependent genes were identified as a result of these efforts. Cycle arrest, DNA repair, death, and senescence are the main biological processes that are connected to the genes that encode p21Waf1/Cip1 and Gadd45 and other Bcl-2 family members.[47,61] The biologically tailored tool most useful positive (most informative positive [MIP]), which searches for novel and instructive positive results, discovers mutations in the tumor suppressor protein p53 that reactivate mutant p53 seen in clinical malignancies. A positive region predicted to be enriched for p53 cancer rescue mutants was identified using MIP, which was used to link all likely single amino acid rescue mutations with p53 mutations reported in patient malignancies in silico.[62]

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5.1 | DNA damage

Humans are constantly exposed to IR, UV light, reactive oxygen species, and other chemical and physical factors that might damage genomic DNA. Surprisingly, mild and severe DNA damage is absorbed into cellular signaling networks in distinct ways, resulting in alternative cell fate decisions.[63,64] DNA damage is a common occurrence that demands a variety of DNA damage repairs to ensure that genetic information is faithfully passed during cell division. Endogenous and exogenous sources can both cause DNA damage.[65] Because of its role in DNA damage response, P53 is known as the "guardian of the genome." The DNA damage response is influenced by the kind, timing, and degree of the stress signal as well as by the cell type. P53 dynamics play a significant role in DNA damage response, and numerous models have revealed a variety of time-dependent p53 profiles. Oncogenic stress activates the ARF tumor suppressor protein, which inhibits MDM2, whereas DNA damage promotes p53 PTMs.[66]

5.2 | Oncogene activation

Oncogenes code for proteins that regulate cell processes like proliferation and apoptosis. Oncogene activation involves genetic changes to cellular proto-oncogenes. A proto-oncogene is a regular gene that, as a result of a genetic change (mutation) that causes an increase in expression, has the potential to develop into an oncogene.[67] The cell acquires a growth advantage as a result of these genetic modifications. In human neoplasms, three genetic routes activate oncogenes: mutation, gene amplification, and chromosome rearrangements. Mutations activate proto-oncogenes by causing structural alterations in their encoded proteins. These changes, which usually affect crucial protein regulatory areas, generally result in the uncontrolled, continuous activity of the mutant protein.[68] Genes that code for proteins known as proto-oncogenes influence mitogenic signaling and signal transduction to influence cell development and differentiation. When a proto-oncogene (or its product oncoprotein) is activated, it becomes a tumor-inducing agent. RAS, WNT, MYC, ERK, and TRK are some of the most well-known proto-oncogenes associated with p53. Another oncogene involved in chronic myelogenous leukemia (CML) caused by a chromosome 9 translocation is the BCR-ABL gene, which is found on the Philadelphia chromosome.[67]

5.3 | Nutrient deprivation

During times of food scarcity (starvation) or oxygen shortage, cells suffer a heavy strain (hypoxia). Similar to persistent nutritional overload, excessive reactive oxygen can harm an organism. "P53 plays a role in metabolism by assisting cells in adapting to and surviving under nutrient-depleted situations such as glucose,[69] glutamine,[70] and serine depletion."[71] By temporarily stopping the cell cycle, P21 (CDKN1A) activation in response to nutritional restriction facilitates protective responses. For example, in reaction to serine deprivation, P21 grants in de novo synthesis of serine into glutathione rather than nucleotide synthesis,[70] and in response to cysteine starvation, P21 helps to postpone ferroptosis.[71] P53 is critical in nutrient restriction because it initiates alternate pathways to ensure cell survival, as well as cell death when recovery looks unlikely.

6 | P53 AS A TUMOR SUPPRESSOR GENE

The p53 gene is a tumor suppressor gene, like the Rb (retinoblastoma protein) the gene that suppresses the growth of tumors. Children with a functional copy of the p53 gene are prone to cancer and a range of early adult tissues generate many different malignancies.[72,73] Li-Fraumeni is a rare disease of cancer. p53 mutations are observed in practically all forms of tumors, contributing to the complex molecular network of events leading to the creation of tumors.[74–76] On human chromosome 17, the p53 genes were identified. The p53 protein is associated with DNA in the cells, leading to an additional gene creating an interacting P21 protein with a cell-dividing protein (cdk2). When p21 connects to the cdk2, the cell cannot progress to the next stage of cell division. The p21 protein is not made available for all cell division as a stop signal because the p53 variant can no longer bind DNA efficiently. Tumors occur in an uncontrolled division of cells.[14,15,77] The application of mice as models for human cancer, using advanced "gender knock-out" procedures, has contributed to the study of the molecular routes of malignant development.[23] The amount of evidence available in human malignancies on all facets of p53 is now immense and shows its essential involvement in cancer pathogenesis. p53 is just one element of a complicated cascade of events leading to tumor genesis.[73,78] p53 is a tumor suppressant and activator functioning Genetic sequence protein.[79] It's called "the guardian of the genome" in cancer biology. The cell cycle, DNA repair, and cell death

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are among other things involved.[80] In human cancer, TP53 is typically identified to be inappropriately mutated, with one nucleotide being replaced by another, as are most tumor-suppressing genes such as RB, APC, or BRCA.[81] Uncontrolled cell development is prevented or damaged DNA is removed when p53 is active and connected to particular DNA sequences in cells with a disturbance in cell proliferation. Because of conformational changes in the p53 protein structure, loss of p53 function induces tumors and results in gene mutation. p53 gene mutations in human malignancies are the most frequent genetic abnormality. All TP53 gene coding exons can have TP53 mutations and, in exons 4–9, a considerable majority code for the protein-binding domain. TP53 mutations are present. Around 30% of alterations occur in six hot areas that occur in nearly all cancer forms (R175, G245, R248, R249, R273, and R282).[82] These mutations result in the accumulation of the mutant p53 protein, which is unable to bind downstream genes and the consensus double-stranded DNA. The p53 protein is released and diffuses into the bloodstream as tumor cells decompose and die. The amount of serum p53 protein is significantly increased in several cancers.[6,17,18]

7 | P53 AND CELL-CYCLE ARREST

Cell-cycle arrest and apoptosis are the effects of p53 activation that are most readily seen. Numerous studies have shown that the actions of p53 cell-cycle arrest and apoptosis are essential in the prevention of tumors. The transcriptional activation of p21/WAF1 by p53 is the primary cause of cell-cycle arrest. Using an unbiased subtractive hybridization method, the p21 mRNA was identified as the first p53 target gene after p53 activation.[83] p21 causes G1 arrest in the cell cycle by attaching to the cyclin E/Cdk2 and cyclin D/Cdk4 complexes.[84] Two distinct locations are where p53 binds. Upstream and downstream of the p21 promoter are separated by 2.4 and 1.4 kb. The 5′ locations on the p21 promoter are one of the strongest p53-binding sites discovered, with a dissociation constant of 5 nm.[85] Recent studies have shown that activation of microRNA mir34a causes p53-induced growth arrest, most likely through suppressing the transcription of several target genes.[86] To preserve chromosomal integrity and enhance the survival of wounded cells, p53 stops cells in the G1 phase and allows time for the repair of potentially lethal DSBs. p53 controls many DNA recombination and repair-related genes as well as a cell cycle checkpoint.[87] When the mitotic spindle is disturbed, the postmitotic checkpoint function of p53 is crucial for preventing DNA re-replication. Normal mouse fibroblasts treated with medicines that destabilize the mitotic spindle, such as nocodazole and colcemid, quit mitosis and enter the G1 phase, where they contain 4N DNA. When a spindle toxin prevents cell division, p53-deficient cells frequently restart DNA replication, resulting in polyploid cells with 8N and high DNA complement.[88] Tumor cells undergo premature senescence as a result of DNA damage, which can continue for weeks after the DNA-damaging treatment is stopped. Several strategies may be used in tandem to maintain senescent cells in a stable cell-cycle arrest. Despite popular belief, p53 inactivation can cause cell-cycle re-entry in senescent cells, contrary to popular belief. As a result, senescence is a distinct type of cell-cycle halt that is extremely stable but not irreversible.[89]

8 | P53 AND APOPTOSIS

The cell cycle consists of many phases governed by several variables. The p53 protein is the most famous negative regulator. Cancers can lead to cancer by mutating or inactivating the p53 gene or by interacting with oncogene DNA viral tumor products.[90–93] These mutations seem to be human cancer's most common genetic modification.[94] Two unique convergent pathways to apoptosis exist in mammalian cells: regulated, intrinsic, mitochondrial, or stress-restricted, BCL-2 and death receiver, also called extrinsic.[95] The loss of activity of the p53 tumor suppressor gene is a frequent and important step in the initiation or progression of many human malignancies.[96] Cell cycle arrest, cell senescence, DNA repair, metabolic adaptability, and cell death are only a few of the physiological processes that are governed by the homotetrameric transcription factor p53's regulation of 500 target genes. It was thought that p53 suppresses tumor formation by inducing apoptosis in nascent neoplastic cells for a long time.[97] p53 is thought to be involved in the process by which cells go from a growth-arrested to an actively dividing state since late G1 is characterized by an increase in p53 protein and mRNA production as well as steady-state levels.[98] Despite readily detectable mRNA expression, unstressed, nontransformed cells have very low (often undetectable) levels of Wild Type p53 protein.[98] The fundamental explanation for this is that the E3 ligase MDM2 targets p53 for proteasomal degradation.[98,99] In response to various stress stimuli, such as oncogene activation, DNA damage, or nutrient deprivation, the levels of p53 protein increase significantly because many signaling pathways that are activated in response to the aforementioned stressors converge on the inhibition of MDM2, while

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others result in modifications in the protein p53 itself.[100–102] The BCL-2-regulated pathway, also known as the intrinsic, mitochondrial, or stress pathway, is triggered by stress conditions like cytokine deprivation, ER stress, or DNA damage, and the death receptor pathway, also known as the extrinsic pathway, which is triggered by ligation of members of the tumor necrosis factor receptor family bearing an intracellular death domain, eventually come together in mammalian cells.[19,103–105] Apoptosis is a morphologically and biochemically unique type of cell death that is triggered by a genetically designed cell suicide mechanism.[106] Caspases are a class of cysteine-containing, aspartate-specific proteases that play a vital role in apoptosis in mammals. Caspases are crucial components of the cell death process in mammals.[107] Multiple apoptotic pathways in mammalian cells have probably developed to allow for the greatest flexibility in cell fate regulation. Understanding these several apoptotic pathways and developing medications that particularly inhibit or activate them will aid in the treatment of a variety of disorders caused by apoptosis abnormalities.[108]

9 | LITERATURE REVIEW

To learn more about how malignant tumors form, cancer researchers spent a lot of time in the 1970s studying how tumor viruses change into cancer cells from normal cells. A few years before its identification in 1979 the discovery of p53 began. Peter Tegtmeyer and his fellow members,[5,109] research into the alteration and regulation of SV40 large T antigen named the A protein, discovered the initial proof of the presence of the protein in the mid-1970s. Anti-SV40-transformed cell extracts are used by Tegtmeyer and his team at New York State University to precipitate SV40-infected cell proteins. Proteins of various sizes were discovered, including one with a molecular weight of 50 kDa. Tegtmeyer's team ignored the 50-kDa protein concerns, most likely because smaller protein variations of large T antigen (94 and 84 kDa) are uncommon. Tegtmeyer's 1972 and 1977 articles,[5,109,110] p53 reports, indicating that later on, 50-kDa protein has been identified as p53.

Lane and Crawford[111] discovered the p53 protein when they explained the role of viruses in cancer through the use of their ability to form a complex with viral tumor T-antigens. p53 was discovered in 1979 as a result of viral analysis and cancer analogy research. Researchers have focused on simian virus 40 (SV40), a tiny DNA tumor virus that is coding two tumor antigens, a small-T-antigen (SV40) and a large-T antigen, to understand more about how DNA tumor viruses might cause cancer. Several investigations have converged to identify tumor antigens, cell protein, and antisera in a variety of cancer-cell p53 proteins. T antigen is required to start and keep the transformed state going. To comprehend the process of the SV40-driven transformation, researchers have been attempting to understand how this T protein interacts with components of the host cell. If rabbit anti-T serum extract is an immunocompromise for the 53K protein, it is present in a T-complex. On the other hand, the 53K and T antigens are both specifically bound by the hyperimmune mouse anti-T serum. The antigens 53K and T both have antibodies and particular non-cross-reactive antigenic determinants. During vivo and vitro testing, they discovered that another batch of serum could bind the separated 53K proteins from SVA31E7 cells. The nature of the 53K protein and its specific binding to the T antigen, as well as the presence of an antigenically similar protein in polyoma-transformed cells, imply that it may play a significant role in controlling the transformed state. Protein is expressed in Daltons (Da). Sulfate-polyacrylamide sodium dodecyl gel (SDS-PAGE) has been frequently used to research proteins in molecular biology for various samples of people and animal cells. Researchers are utilizing a marker that holds proteins of predetermined sizes to identify the protein size of the samples. This parameter assists scientists in classifying and sometimes calling proteins. This was the name of the p53 protein since it weighs 53 000 Da.

More evidence of the presence of this mysterious protein was soon found, and investigations in 1979 have been flourishing. In 1979, DeLeo et al.[3] discovered a novel antigen in sarcoma and other altered mice cells that were created chemically. p53 reactivity was tested in antisera against structural antigens of leukemia murinus, and all of its antisera and antisera heterologous; besides Meth A antisera, only CMS4, which is a similarly produced anti-p53 sarcoma, was shown to be anti-p53 activity. All investigated cells of the mouse, including chemically produced sarcomas, leukemia, endogenous fibroblasts, SV40 transformed cell, and murine sarcoma virus, be p53. p53 was found in immunological research of cancer due to its immunoreactivity with the tumor antisera. The most basic approach to assess whether the transformation of viral code for p53 is in vitro synthesis of viral gene products and the presence of p53 in transformed cells, such as MuSV and SV4O.

Kress et al.[4] reported similar results on the existence, by several SV40 transformed cells, of two new protein classes (so-called "*super-T proteins*") of molecular weight greater than the large T antigen (86 000) 86 kDa. The other protein (termed "*55K proteins*") with molecular weight ranging from 50 000 to 60 000 was presented in all

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SV40-transformed cells. In many cell lines, it has been reported that in addition to large T and small t antigens, an additional protein can be found, immunoprecipitated from different SV40 transformed lines, using an anti SV40 tumor serum, which is a protein of approximately 55 kDa that links to large T antigens in diverse cancerous cell types. It can also be found in many SV40 transformed cell lines. 55 K proteins are recognized as fatal antigens, which are not usually expressed in the types of adult cells, but are depressed during the transformation process caused by viruses.

In August 1979,[89] a similar finding from a 55 kDa protein in different types of malignant cells was reported in two more experiments. In 1979, the 40 altered mouse cell antigens of a simian virus were found by Chang et al.[89] in addition to big (94 000 Da) and minor tumor antigens (20 000 Da). Some of these converted SV40 mouse cells found one molecular protein of around 130 000. In all virus-transformed cells, the other, known as Tau antigen, has a 56 000 molecular weight, according to the electrophoresis of acrylamide gels. In all cells converted by viruses tested using electrophoresis in acrylamide gels, the other protein, termed Tau antigen, had a 56 000 molecular weight which strongly indicated that the protein had a virus coding. The presence of Tau antigens in cells transformed by SV40 can be explained in different ways. Tau antigens may be fully viral coded first and substantially modified after synthesis, leaving only 2 methionine-tryptic peptides unmodified. The alternative hypothesis that the T and Tau antigens are translated from the previous SV40 RNA into two independent frames of the codon is unlikely to occur in two of the three possible readings during the most recent transcripts.

Melero et al.[112] found comparison in cells transformed or infected with DNA tumor virus SV40, which are addressable with an antiserum T. A wide range of models for carcinogenesis and cell transformation in culture was investigated for the DNA tumor virus SV40, as well as for the related papova viruses polyoma and bk virus. During this study of these proteins, the authors tried to find out the answers to the following questions:

1. Are they host or viral products?
2. Where do they occur in the cell?
3. Relationship between the primary structure.

Here they used a technique, Polyacrylamide gel electrophoresis is one of the powerful tools for analyzing RNA samples widely used in biochemistry, for protein purification from various mammalian species, and they finally conclude that the 48 000 and 55 000 molecular weight proteins which share similar functions and products of different cellular genes, and 48 000 protein is host-coded.

In 1980, Rotter et al.[113] identified a new protein p50, present in regular and non-A-MuLV induction cell levels but high in the majority of A-MuLV-producing tumor cells. Mice are typically exposed in the case of A-MuLV-induced lymphoid tumor cells to P50 antibodies, a host-coded protein. An antibody in Sera was discovered from the highest-level Sera tumor mice of mice that refuse to receive their tumors and that immunosuppressed a special protein, A-MuLV transforming Balb/c cells from 35S′ methionine. This protein was a 50 000 molecular weight protein called P50, not an A-MuLV-specific protein previously found (P120). It was shown that p50 was a host cell protein whose concentration was specifically enhanced with the A-MuLV process at low levels of cells and transformed by agents other than A-MuLV and was not a Moloney MuLV protein. The three proteins (55 000–60 000 molecular weights),[4] host protein (48 000–50 000 molecular weights),[112] and 50 000 molecular weight p50 are highly related and most likely identical, according to preliminary studies. p50 is thus raised in a range of tumor cells and may play a role in tumor cell malignancy generated by a variety of carcinogenic stimuli.

Rotter et al.[114] studied several cell impacts, including Abelson morning leukemia (A-MuLV), Simian cells (SV40), and 50 000 molecular weight cell protein chemically modified cells, in 1981. The serum p50 protein, increased to chemically modified and A-MuLV cells and highly bound to T-antigen in SV-40-transformed cell extracts, was identified. A cocapping experiment has indicated a weak connection of the two proteins, indicating but not showing any association signs with the immunoprecipitation (IP) experiment that the surface P50-related protein is linked partially to the a-muLV protein. Here they find that p50 is an uncommon protein and has only a few hours of half-life in the cell and interacts weakly on the surface of the cell.

The major finding of the immunogenic p53 protein in 1983 by Crawford et al.[115] Antibodies against the host protein p53 have been discovered for patients with primary or secondary breast cancer. The finding of anti-p53 activity in patients with cancer will confirm the idea that a change in the number or type of p53 might in some conditions involve in vivo malignancy. Immune response genes will also influence the synthesis of antimicrobial compounds, and many people can simply not respond to p53 as an antigen. It will be interesting to observe if sera with anti-P53 activity was taken from those who are more prone to become stage I and step III and develop lung secondaries later. Three of the five patients with anti-p53 activity in their sera when their primary tumors were discovered had recurrences, with two of them having lung secondary tumors. There are a few probable

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reasons why we didn't observe activity in the sera of many cancer patients. After all available p53 was removed by thorough precipitation with PAb421, an extract of SV40-infected CV-1 monkey cells was used as a free source of SV40 large-T for IP to investigate anti-T efficacy. This demonstrated that the two sera were targeting the SV40 large-T antigen with their activity. At the same time, it was found that a strong anti-p53 serum had no detectable anti-SV40 large-T action. This is significant because it indicates that anti-p53 and anti-SV40 large-T activity were the sources of the anti-p53 activity in this positive serum.

The immunological specificities of the anti-54KD monoclonal antibodies and partial peptide maps for both adenoviruses and the SV40 tumor antigen are used to identify the 54KD protein, investigation done by Peter Sernow and other, researchers[116] in 1982 showed that there is a relationship in adenovirus-transformed mouse cells between the Elb58KD tumor antigen and a 54-kDa cellular protein. Immunological speciality for anti-54KD of monoclonal antibodies and incomplete peptides of 54KD of protein related to adenovirus and tumor antibodies will be used to determine protein identification of 54KD. El b58KD protein is coimmunoprecipitated with a monoclonal antibody specific for the cellular 54KD protein from transformed cell extracts. The physical connectivity of the adenovirus and the transformed cell SV40 tumor antigen, which are critical in transformation, are solid evidence of the same mechanisms or roles in these two separate viral proteins. The discovery that the Ad2 tumor serum, containing 58KD-protein antibodies, but not 54 KD of protein, immunoprecipitates a 58KD-54KD cell complex proves that the Elb-58KD tumor antigen is physically coupled with a 53KD cell protein in viral mice cells.

Benchimol et al.[117] developed simple and rapid radioimmunoassays (RIAs) for the cellular protein p53 that are both sensitive and quantitative. Therefore, until the immunofluorescence is proven to be caused by the availability of a cross-reactive species in the cell, reports on p53 levels based solely on immunofluorescence data are questionable. The rate of synthesis (KS) of a protein can be determined by pulse-labeling metabolically active cells with radioactive amino acids and isolating the radiolabelled protein using IP and gel electrophoresis. The relatively stable equation $P = KS/KD$ requires good knowledge of the rate constant of degradation (KD) to extrapolate this value to the protein's steady-state level (P). In reality, detecting human p53 is just as easy as detecting mouse p53. The mouse p53 assay may not be as sensitive to the p53 protein's condition as the human p53 assay is. The p53 levels found in several cells employing the designated RIA varied by more than 100-fold. The highest amount of p53 is found in SV40-transformed cells, although other transformed cell lines that are not SV40-transformed also have very high levels. As seen by the measurements of p53 levels obtained in MCA (chemically transformed) and CQ15 cells, the RIA can assess free p53 without the SV40 big T antigen. According to tests using IHC and immunoassay, human malignancies and cell lines exhibit higher concentrations of the p53 protein.

In 1982, Mercer et al.[90] proposed that p53 be involved in cell growth regulation. In developing normal cells, but in fewer numbers, P53 was also identified. When nondividing Go, cells are induced to proliferate by the presence of concanavalin A, p53 is produced in the same lymphocytes. P53 has been postulated that the mammalian cell cycle is regulated because it is related to cellular proliferation and transformation. In the nucleus of the quiescent 3T3 cells, Preimmune IgG microinject did not diminish serum-stimulated DNA synthesis, whereas monoclonal antibodies were used to prevent SV40 T antigen. Although immunofluorescent and autoradiography experiments do not demonstrate that p53 is involved in the start of cell propagation, they suggest that it is a major ingredient in the cell change from rest to a growing status.

The fact that the steady-state growth in p53 protein levels in the year 1984 is attributable to an increase in p53 production instead of just the build-up of the protein p53 in nontransformed cells is associated with the control of cellular proliferation is demonstrated by Reich and Levin[118] in 1984. Here they checked the steady-state level of p53 by analyzing it after adding serum at different time slots. The p53 was immunoprecipitated from cellular extracts, electrophoresed in polyacrylamide gel, and changes were noted in the level of p53. Just before entering into the S phase, the levels had increased by 10–20-fold after 6 h of the experiment by adding serum. The p53 was then identified using the western blot technique. When it was thought that the stability of the protein would increase in late G1, pulse-chase tests were utilized to test the p53 half-life. It was observed in earlier investigations that the DNA synthesis inhibitor did not alter the pattern of rising p53 and mRNA levels. Therefore, the time of p53's increased expression may be related to the protein's functional activity.

p53 has a major impact on changing the control of cell division in transformed states. It plays a substantial qualitative or quantitative function in the change of cells from a resting to a growing stage, in other words. In 1985, Mercer et al.[119] examined the expression of the p53 protein in human peripheral blood cell lymphocytes under various circumstances following mitogenic

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stimulation under various cultures to investigate the part performed by p53 in the remainder of the normal cell transition. The objective was to comprehend the function of the p53 protein in the control of mammalian cell development. The new findings from IP and autoradiography experiments are as follows: (1) in human lymphocytes, inducing the p53 protein is correlated, instead of with the cell size growth, in the extent of the cell DNA synthesis[7]; (2) interleukin-1 increases the p53 protein synthesis and(3) monoclonal antibody inhibits the accumulation of p53 protein in the interleukin-2 reception (antiprotein). (4) p53 protein expression is inducible by various mitogens of different types of cells (this term is used herein in one of its accepted applications, that is, as indicated by synthesis level or the gene product quantity). There is no doubt in our immunoperoxidase tests that p53 is located in the nucleus of normal human cells.

Oren and colleagues[120,121] express the numerous pathways that contribute to higher p53 levels which could help to clarify the link between this protein and metamorphosis. Nonetheless, the lack of a suitable nucleic acid probe severely hindered research in this area. To address this issue, they were trying to create a cDNA clone of the SVT2-cell from p53 mRNA, a mouse-transformed SV40 cell line. This paper presents the isolation and identification of such a cDNA clone, dubbed pp53-208, as well as its preliminary characterization. This paper discusses the creation of pp53-208, a plasmid containing sequences found in the mRNA of the murine p53 cellular tumor antigen. The availability of a p53 cDNA clone now allows researchers to tackle this issue more directly, as well as examine various cells and tissues for p53 mRNA overproduction. The detection of variations in p53 gene expression with the help of the IP technique and the identification of variables influencing those changes should aid in determining the precise connection between p53 and cell transformation.

In the same year 1984, Matlashewski et al.[122] used the isolation and identification for cDNA clones obtained and analyzed for human cellular tumor antigen while in the previous paper they used murine p53 cellular tumor antigen. Using the human p53 cDNA clone, the genomic architecture of the p53 gene in transformed and healthy cells has also been studied. By using hybridization selection techniques, it was demonstrated that this clone could preferentially bind p53 mRNA. A variety of human transformed cells were examined for the presence of p53 mRNA using a nucleic acid hybridization probe made from plasmid 102. According to Northern blot analysis, p53 mRNA is present in all cell lines.[122] The primary amino acid sequence of p53 from the magnetic mouse and the human amino acid sequence predicted for cDNA are relatively similar (84%). The human p53 cDNA has an Alu repeat region but the mouse p53 cDNA does not. Although the coding sequence of the p53 protein is very similar, the Alu sequence disrupts a region of human p53 mRNA that is analogous to mouse p53 mRNA. Additionally, human p53 mRNA has a significantly different structure than mouse p53 mRNA. Although the untranslated region of human p53 mRNA is different overall, the 3′-untranslated region of mouse p53 mRNA codes for roughly comparable proteins.

The role of p53 in both ordinary and changed cells was given in the work in 1985[123] by Bienz-Tadmor et al.[124] with the use of clone isolation. The cDNA cell library has produced three clones of the human tumor antigen p53. pR4-2 is one of the clones that has been detected and has all the acid sequences necessary to code the whole coding region with human p53, but not a complete copy with the human p53 mRNA. The p53 mRNA is approximately 2500 nucleotides long, according to Northern blot studies, but only 1760 base pairs for pR4-2. p53 cDNA was extracted, marked, and sequenced by the p53 cDNA clone of the Human cells A431 with the entire length coding potential of p53. The clones reported here will help us better understand the role of p53 in both normal and altered cells.

Eliyahu and colleagues[8–10] report on the cooperation in cells' transformations between p53 tumor antigen encoding and Ras. Protein p53 has a wide array of generated cell types and is converted by numerous species, including SV40, adenovirus and Abelson virus, and several cells chemically modified. In certain untransformed cells, some embryonic organs possess considerable p53. It appears that p53 plays a significant role in carcinogenesis. The calcium phosphate co-precipitation method was used to show that there is a clear connection between tumourigenicity and a high p53, and the p53 coding gene can be a cause of conversion of normal fibroblast into tumor cells and of cell immortality and neoplasty sensitivity to Ras oncogene.

Jenkins and colleagues[10,125] discussed the topic of immortalization of fibroblast by the p53 oncogene. The Ras oncogenes' encoding gene, p53, can contribute to cellular longevity, susceptibility to neoplastic transformation, and the conversion of healthy fibroblasts into tumor cells. Malignant primary-service transformation requires a minimum of two separate mobile behavior modifications. First of all, a chemical carcinogen or a gene for replicated cancers, such as the big T polyoma, the early adenovirus 1A area, or the avian myelocytomatosis virus v-myc tumor gene, may lead to the immortality of the cells. Experiments demonstrate that an increase in mouse expression p53 gene can easily

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immortalize early-cast rat embryos fibroblast, that is, p53 (mutant) immortalize the rat's chronic cytes and embryo fibroblast in the absence of a potent heterologous promoter and/or enhancer controlled by an indigenous promoter. Western blot and metabolic labeling techniques were used to confirm the expression of murine p53 and p44 in the developed rat cell lines. Their findings show that, in the absence of strong foreign promoters and/or enhancers, enhanced expression of the mouse p53 gene, driven by its native promoter, can easily immortalize early-passage rat embryo cells.

Wolf and colleagues[126–130] described p53 as a gain in oncogenic function due to p53's mutant and metastatic potential. The results that have been seen support the hypothesis that malignant changes are multistage. This process appears to have halted in the case of L12 variants without the presence of the p53 protein. A single example of rapidly proliferating cells that lack p53 protein is L12 Ab-MuLV-transformed cells. The majority of the clones had stably absorbed p53 DNA sequences, according to Southern blot examination. These cells probably generate only a tiny number of 53 modules that cannot be detected in an experiment but are enough to meet the needs of the cell cycle. There seems to be a quantitative major difference among transformed cells; transformed cells produce p53 more than untransformed cells. This quantitative difference could be due to posttranslational regulation mechanisms, according to some theories.

The p53 cellular tumor antigen, which has long been recognized to be overproduced in a variety of neoplastically alter cells, has discovered by Eliyahu et al.,[128] which showed that the p53 plays a direct role in transformation.[128] Proteins were immunoprecipitated and examined by SDS polyacrylamide gel electrophoresis, which demonstrated that p53 can assist in activating HA-RAS by transforming secondary rat embryo fibroblasts into significantly changed tumorigenic cells. Furthermore, the level of P53 overproduction may be connected to the tumourigenic efficacy of established lines. The data provided here strongly suggest that increasing the amount of P53 produced is sufficient to establish sidelines to become tumourigenic.

Pohl et al.[129] addressed the increases in the level of p53 in the experimental metastatic capacity of murine carcinoma cells. After transfecting p53 into a mouse bladder cancer cell line with a low metastatic capacity, the authors found that these cells had a significantly increased ability to colonize and multiply in syngeneic recipients' lungs after intravenous inoculation. Numerous altered cells showed higher p53 levels in the culture, indicating that p53 is connected to cancer by being related to malignancy. The SDS-polyacrylamide gel electrophoresis profile of immunoprecipitated p53 protein revealed increased metastatic capacity, which might be connected to an increase in p53 protein levels. The role of p53 was examined in a developed cell line of murine bladder cancer. This cell type has considerably boosted its colonization and its spread to a mouse bladder cancer cell line with minimum metastatic capacity in the syngeneic lungs following the intravenous injection of p53. After genomic p53 transfection, new genes p53 were integrated and expressed.

Wolf and colleagues[131,132] explained how the insertion of Moloney virus-like sequences in the 1st p53 intron in Ab-MuLV-transformed cells inactivated functional p53 cells. The results that have been seen support the hypothesis that malignant changes are multistage. This process appears to have halted in the case of L12 variants without the presence of the p53 protein. A single example of fast-growing cells lacking p53 protection is L12 Ab-MuLV transformed cells.[131] The p53 is essential to the transition to a neoplastic condition of in vivo cells and there is a change in the expression of the cell's p53 genes such as the rearrangement of the Moloney leukemia gene or the Friend leukemia virus. p53 is transformed by different isolates of the buddy leukemia virus and it is shown here. Western blot technique is used to analyze the tumors which are formed by p53-negative and positive cell lines, which resulted in no p53 protein being found in tumors formed by cell lines DA10, CB3, CB5, and DP16-1, while because CB7 and DA9 cell lines p53 was detected in tumors.[132]

Lu and colleagues[21,133–136] are seeing a significant rise, following DNA damage, in transcriptions in wild-type cells p53 and cell cycle arrest of reporter genes having p53 binding sites in their sponsors. In the nucleus of mammalian cells, p53 protein accumulation was reported for damage to the DNA caused by UV, X-ray, or restriction enzymes. Using quantitative and qualitative immunochemical techniques, they studied the kinetics and amplitude of the p53 response to a variety of triggering events. The build-up of 53 can be induced by DSBs caused by the introduction of a restriction endonuclease into cells, as well as DNA lesions caused by UV and X-ray, but not by heat stress. The results show that enhanced p53 expression is associated with UV radiation and the UV-mimetic chemical carcinogen 4NQO reactions in mammalian cells. The results presented here suggest that p53 expression is not only a reflection of cellular proliferation but is also involved in some aspects of DNA synthesis. They demonstrate a relationship between elevated levels of p53 and cell responses to a class of DNA-damaging agents whose immediate effects do not include stimulation of cellular growth. In conclusion, they found that TPA-induced

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myeloblastic leukemia and normal human hematopoietic cells both boosted the growth of endogenous, wild-type p53 protein. These results suggest but do not prove, a causal relationship between endogenous p53 expression and the cessation of cancer.[134]

Isobe and colleagues[137–140] give rise to the conclusion from the laboratory's studies that the p53 gene indeed maps to chromosome 11 in mice and chromosome 17 in human. They established a strategy that differentiates p53 from mouse to hamster from Chinese by comparing certain anti-p53 monoclonal antigenic determinants.[139] The difference in antigenic determinants between mouse and Chinese hamster p53 is a good method for locating the p53 chromosomes in mouse x Chinese hamster cell lines. Only the hybrid cell line 22 yielded the mouse 16-kb region matching the active gene. They infer that the polypeptide p53 gene should be situated on chromosome 11 of the mouse because it is the only cell line to carry it.[138] Through the use of Southern filter hybridization of DNA by human rodent hybrids, the p53 gene has been discovered in the short arm of human chromosome 17. Chromosomal localization could help researchers better understand the interaction between p53 and other human cellular genes, as well as its role in cancers linked to certain chromosomal rearrangements.[140]

To understand the role of p53 in the malignant cell process, Wolf and colleagues,[141,142] comprise comparative investigations between p53 cell producers and nonproductive variations, and human malignancies with p54 genetic arrangements were identified. The oncogene family members were split into two basic categories based on functional and structural criteria. p53 is most probably one of the members of the family of nuclear oncogenes, and, in line with recent discoveries from multiple experimental systems like southern blot analysis and RNA blot analysis, are complemented by the action of oncogenes in the second category like the myc, myb, and elA proteins. The human cell line of the malignancy HL-60 is remarkable in its absence. The p53 gene in these cells is extensively changed according to the HL-60 DNA analysis.

Molecular structural analysis of the p53 gene in patients in either the accelerated phase or blastic crisis revealed a high rate of gene rearrangements in people with Philadelphia chromosome-positive CML.[141] As a result, they believe that one of the pathways leading to the acute phase of CML is linked to mutations in the p53 nuclear oncogene.[141] The chronic stage affected 39 of the 53 patients, while the blastic crisis or accelerated stage affected 14 of the patients. The disease's stage was established using clinical, hematological, and cytogenetic criteria. In the current study, they examined the p53 gene structure in CML patients and found that during the blastic crisis, 30% of patients had p53 gene rearrangements.[141] p53 is a malignant nuclear protein in many molecular weight molecular tumor model systems of 53 kDa. p53 gene rearrangements have only been discovered in osteogenic sarcomas in 134 human carcinomas (three of six osteogenic sarcomas evaluated), Masuda et al.[143] in an examination of 134 human sarcomas, leukemia, and peripheral blood lymphomas.

The alternative splicing discussion by Wolf and colleagues[144–146] produces immunologically different p53 molecules. In this publication, they provided the first experimental proof that variably spliced p53 exists in normal cells of many tissues and is not a transformation-specific event type. p53 tumor antigen is most likely a member of the nuclear oncogene family, according to recent discoveries from numerous experimental methods like RNA blot analysis and Southern blot analysis, and it complements the activity of oncogenes in the second category, like myc, myb, and ElA proteins. Both normal epidermal and carcinoma cells contain around 25%–30% of the routinely spliced p53 RNA of the alternatively spliced tumor suppressor gene p53 which has 96 extra nucleotides generated from intron 10. Results demonstrate that the single mouse functional p53 gene codes are produced by separate splicing mechanisms for at least two p53 mRNA types. Wolf et al.[144] described their approach for determining the function of p53 in the process of malignant cell transformation by comparing studies between p53 producers' and nonproducers', and they found that p53 was genetically altered in human malignancies.

The studies of Delphin and colleagues[12,147–159] show that serine phosphorylation (casine kinase II, DNA-dependent protein kinase, casein kinase such as kinase, protein kinase C, p34cdc2, MAP kinase, C-jun kinase, raf) and ribosomal RNA covalent attachment modifies the p53 protein after translation. The tumor suppressor protein p53 is closely adjusted and phosphorylated by many protein kinases at various locations. Kinases of mitogen-activated protein (MAP), a protein kinase implicated in a variety of mitogenic and differentiating signals, were used to study the phosphorylation of p53. p53 is reported to be upregulated in cells changed by multiple agents, including tumor DNA viruses such as SV40, adenovirus type 2, and the Epstein–Barr virus, as well as various spontaneous tumors. Although evidence exists to suggest that phosphorylation plays a role in enzyme function modulation, nothing is known about how phosphorylation affects the function of nuclear oncogene products. They aimed to map the phosphorylation sites of p53 to better understand how phosphorylation affects its biological activity. NIH 3T3 (a continuous line of strongly contact-inhibited cells from the

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NIH Swiss mouse embryo) is transformed into the normal and SV40 (serine phosphorylates) are established in p53.

In p53-plus-ras-transformed cell lines, immunological evidence indicates that the association of p53 with a heat shock protein, hsc7O, Mdm2, WT1, E6-AP, TBP, RPA, SP1, CBF, XPD, TAF, 40,60, spot1, TAFn31, p62, XPB, abl which were suggested by Pinhasi-Kimhi and colleagues.[12,123,160–179] Heat shock proteins are a varied group of proteins that are commonly found in normal cells at low concentrations. In response to physiological stress, the synthesis of these proteins increases quickly, and the pattern of total cellular protein synthesis is significantly changed. At least three proteins with at least 80% amino acid homology make up the hsp70 family of proteins found in human cells. These proteins are referred to as hsp70, hsx70, and hsc70. With a mutant p53 protein that binds to hsc70 proteins more easily than the normal p53, the Ras oncogene appears to transform rat cells more effectively. These findings imply the existence of an active p53-hsc70 complex. Proteins with structural changes may be recognized by Hsc70, a heat shock response protein that travels to the nucleus.[162] They also learned that S100b, a member of the S100 protein family involved in cell cycle progression and cell differentiation, interacts with p53 in a calcium-dependent manner and that this connection prevents p53 from being phosphorylated by protein kinase C in vitro.[12]

Their findings show that human MDM2 binds to p53 in vitro and is genetically changed in a considerable proportion of soft tissue and bone sarcomas. Another intriguing link between viral carcinogenesis and the naturally occurring genomic changes that underpin sporadic human cancer is MDM2 amplification.[163] Human p53's Transcriptional Activation Domain Interacts Directly with the TATA Box-binding Protein.[167] The human p53 tumor suppressor gene produces a transcriptional activator in both yeast and mammalian cells. In this paper, we show that wild-type human p53 binds specifically and directly to yeast TBP using protein affinity chromatography.[170] Numerous transcription factors can start DNA replication. Affinity chromatography (RPA) has shown that the acidic transcription factor fields of VP16, GAL4, and p53 each preferentially bind with human and yeast replication factor A.[171] RPA is a type of DNA-binding protein that is required in many circumstances for DNA replication to begin in vitro. The link between transcriptional activator proteins and RPA may be necessary for DNA replication to start because VP1 8 mutations impact both its capacity to encourage polyomavirus DNA replication and its capacity to bind RPA.

The most recent genetic alteration in a human's genes has been investigated by Braithwaite and colleagues.[17,169,180,181] p53 (mutant) protein is a less efficient antigen-SV40 T binding protein and does not affect SV40 viral replication, while p53 (wild type) protein limits the antigen-SV40T binding alpha-polymerase DNA binding and the ATP-dependent helicase activity. In mice, the tumor-suppressor gene p53 is converted into a dominant transforming oncogene that promotes full transformation in conjunction with the active Ras gene and immortalizes primary cells. They believe that PAb240's IP of p53 in mice and humans represents the diagnosis of a mutation, with the many mutations of sites that change p53 from a recessive to a dominant oncogene having a similar structural impact on the protein. The SV40 T antigen and the adenovirus behave exactly like p53 mutants because E1b binds to both wild-type and pseudo-wild-type p53 molecules.

Prideaux et al.[182] suggested that the internal deletion or retroviral insertion inactivates the p53 oncogene in erythroleukemic cell lines generated by the Friend leukemia virus. The Friend leukemia virus has mutated the p53 gene of a vast number of erythroleukemic cell lines produced through the spleens of mice. The rearrangements in two murine erythroleukemic cell lines produced by Friend leukemia virus, DP20-1, and CB3, that contain a rearranged p53 gene and fail to express p53 protein, have been molecularly described. A molecular clone of the altered p53 gene in a second cell line, CB3, indicated that the p53 gene in this clone has a significant deletion, resulting in the loss of coding regions between exons 4 and 8. The significance of these findings in the multistage Friend virus-induced erythroleukemia.

Squamous-cell carcinoma had a higher rate of loss of heterozygosity than adenocarcinoma. The elimination of DNA sequences in 8 out of 9 heterozygous squamous cell carcinoma (SCC) cases from chromosome 17's short arm (D17S2 locus) was observed, and only 2 out of 11 heterozygous adenocarcinomas. In the bands, 17pl2 to 17pl3.3 with 20 chromosomal 17p markers, the usual area of detection for these malignancies is seen. The transformation-associated protein p53 encoding gene is located in this location, that is, losses in human cancer of chromosome 17p heterozygosity are often observed. In CRC, the deletion of Chromosome 17 and the p53 gene was usually observed by Bakers et al.[183] Small-cell carcinoma and adenocarcinoma changes on chromosome 3p can be strongly influenced by chromosome changes 3p, 13q, or 17p. Loss of heterozygosity on chromosomes 3, 13, and 17 in small-cell cancer and on chromosome 3 in lung adenocarcinoma. Instead of using cultured cell lines, they used RFLP analysis to gain definite information on chromosome aberrations in fresh lung cancers

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Furthermore, this study will determine whether chromosome 3p deletion has common pathogenetic processes in the formation of SCC and ADC lung cancers, which are histologically distinct, Yokota and colleagues[184,185] give rise to the conclusion that alteration in gene structure during colorectal carcinoma development. In 172 colorectal tumor specimens of varying phases of neoplastic growth, they identified 4 genetic changes. The statistics show chromosome 17p in breast cancer families as a potential linkage research applicant. In 1988, Mackay et al.[186] documented the large losses in the short arm of chromosome 17 in breast tumors. The amplification of the Erb B2/neu oncogene and related sequences on 11p, 13q, and 17p are now strongly implicated in breast tumors. A variety of chromosome deletions, translocations, and entire losses were seen in near diploid primary breast tumor cells, with chromosome 1 and the long arm of chromosome 11 at q13 being particularly often affected.

The presence of this modification in allele loss is a regular occurrence in chromosome sections harboring recessive oncogenes, according to Ahuja and colleagues.[187–189] Lung cancer patients frequently lose heterozygosity on 17p. Furthermore, when comparing lung cancer cell lines to normal lung cells, p53 messenger RNA expression was found to be very low or nonexistent. Allele loss for chromosome regions 3p and 17p in SCLC and non-SCLC suggested a search for anti-oncogenes in these chromosome loci. According to cytogenetic, RFLP, or both analyses, all SCLCs and six of the non-SCLCs have 3p deletions or loss of heterozygosity. In addition to understanding the molecular mechanics of lung cancer starting and advancement, further study into how these injuries are reversed and whether their correction restores the malignant phenotype could also generate improved prevention, diagnosis, prognosis, and even treatment techniques.

The results of numerous experiments, given by Finlay and colleagues[22,190,191] showed that the p53-hsc70 complex may be involved in the regulation of p53 level in transformed cell lines. Rat cells are modified when the Ras oncogene is combined with p53 cDNA from methylcholanthrene-transformed cells, and this results in a protein that binds to heat shock proteins.[191] These quantitative differences may be caused by the complex p53-hsc70 increasing the mutant p53 protein's half-life, and p53 qualitative modifications may result in a quantitative rise in p53 levels. The formation of the Meth A tumor was linked to abnormalities in the p53 gene. Exogenous p53 and its endogenous counterpart form a physical combination in cells over-expressing a transfected mutant p53, presumably leading to the latter's stability.[190] In rat embryonic fibroblasts, P53 does not work with Ras-oncogene; this emphasizes that in earlier experiments mutant p53 has been employed.

The complexes with indigenous proteins p53 were previously shown to produce mutant p53 proteins. To control p53 levels in transformed cell lines, the p53-hsc70 complex can play a function, according to various researchers named Finlay and colleagues.[22,125,192–194] As a result, cellular transformation has been linked to an increase in p53 levels. Several investigations have found that higher amounts of p53 cause alterations in the phenotypic of normal cells; overexpression of p53 caused rodent cells to be immortalized. IPs were done on a variety of cell lines that included well-characterized wild-type and mutant p53 molecules. There is no transformant when p53-wt and an active ras gene are cotransfected into primary rat embryo cells. According to recent findings, transforming proteins do not encode the proto-oncogene p53's wild-life sequence (p53-wt). The misinterpretation of p53's transformative role is caused by several p53-cDNAs and at least one p53 genomic clone, which all have mutations that activate the Ras p53 protein.

Lavigueur et al.[34] was examine the participation of the p53 gene in vivo oncogenesis. The study examined the use of transgenic mice containing the genomic fragments p53 of a mouse. Transgenic p53 (mutant) increases transgenic mouse tumor frequency. Inactivation or alteration of Friend Erythroleukemia in vivo p53, a recessive oncogene, may offer an advantage in selective growth to the erythroid target cells infected. The western blotting approach was employed in the majority of tissues to find elevated p53 levels. These findings show that p53 oncogene disfiguring alleles have an in vivo oncogenic potential and that different cell types have different susceptibilities to the transformation of p53-media malignancy. In summary, multiple recent discoveries suggest that p53 is a recessive oncogene.

Previous studies showed allelic suppression of chromosome 17's short arm in about 75% of colorectal carcinomas. Baker et al.[6] concluded in 1989 that chromosome 17p deletions in human cancers are linked to p53 point alterations, and that genetic data indicates that p53 is a tumor suppressor gene. Twenty DNA probes that search for chromosome 17p RFLPs (restriction fragment length polymorphisms) were used to analyze the patterns of allelic loss in CRCs. Based on their hybridization to human-rodent somatic cell hybrids including portions of chromosome 17p, these probes have been linked to seven distinct sites on chromosome 17p. The most prevalent region of deletion was found to comprise 20 17p chromosomal markers in bands 17pl2 to 17pl3.3. DNA was compared to the DNA of the normal

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colonic mucosa with a total of 58 cancer specimens. Allelic losses were assessed in the absence of DNA from one of the two alleles found in normal cells. Seventy-seven percent of the tumors had at least three markers with allelic loss. They were able to identify a common deletion area on chromosome 17p by studying eight cancers that kept heterozygosity for some but not all markers. Even though the p53 gene is thought to be an oncogene, various investigations have revealed that the normal p53 gene may have suppressor activity.

Takahashi and colleagues[188,195] gave proof of the presence of this alteration in allele loss is a common occurrence in chromosome areas with recessive oncogenes. In a variety of human malignancies, the p53 gene is frequently altered. p53 mutations arise in carcinogenesis and are the primary effects on the growth of the tumor. Allelic deletion of chromosome 17p has been found in at least 60% of colon, breast, lung, ovaries, cervix, bone, and bladder tumors, as well as at least 30% of brain tumors, according to previous cytogenetic and RFLP research. According to their research, the majority of malignancies with 17p allelic deletions have mutant p53 genes, and some tumors without these deletions also have mutations of a similar nature. For sequencing analysis, 19 tumors with allelic deletions of chromosome 17p were chosen. Thirteen tumors contained a single mutation of missense, two tumors had two mutations of missense, one tumor had a 293-codon frame-shift mutation, and three were unmutated. Mutations in the p53 gene occur during carcinogenesis and drive tumor development via a dominant-negative effect. The deletion of a chromosome 17p allele has been demonstrated to be related to tumor progression in the human host.

Recent evidence reveals that loss of normal expression in p53 is prevailing in certain forms of malignancy. This loss could lead to selective growth advantages for altered cells. Diller and colleagues[91,92,193,194,196–201] concluded that p53 works as a tumor-suppressing gene by inhibiting human cell proliferation in vitro and/or in vivo. Expression of the wild-type p53 may be tumor-selective. P53 inactivation can play a significant impact on malignant tumor development. Previously, mutant protein p53 had become complexes with physiological protein p53.[196] In p53-level regulation, the p53-hsc70 complex could be used in cell-transformed lines. P53 gene changes in colorectal carcinomas are common and wild p53 are frequently removed at the same time.[193] These findings show that the wild gene can prevent the vivo-derived mutation of the p53 gene product's single conservative amino acid replacement, which can result in the in vitro development of human colorectal cells.[197] The gene encoding p53 is a 53 kg cell-based tumor cell, which implies that this gene plays a major role in human oncogenesis.[197] A human cell line of osteosarcoma Saos-2, which lacks endogenous p53, was used to reproduce the gradual mutation or loss of both p53 alleles throughout carcinogenesis. The significance of the p53 gene control pathway in established osteosarcoma cell lines was discussed by Diller et al.[91] Our results reveal striking differences in the effects of mutant and wild-type p53 genes on osteosarcoma cell proliferation. A new technique for identifying the regions of the cell cycle that the wild-type p53 gene controls are being developed using flow cytometry on cells that have been transiently transfected.

Malkin and colleagues[74,75] reported on germline p53 mutations detected in families susceptible to carcinoma and were published in 1990. Many inherited and spontaneous forms of malignancy were related to tumor suppressors, which appear to inhibit normal cell proliferation with normal function. This impact is affected by several mutations in the suppressive p53 gene, which occur in a range of malignant malignancies. The deletion of one allele and the substitution of a single nucleotide in the other allele are the most common mutations. In a family with Li-Fraumeni syndrome, they have now looked to the gene p53, a rare autosomal dominant disorder defined by multiple mesenchymal and epithelial neoplasms. They looked at 18 people from six generations of the Li-Fraumeni family who had previously been diagnosed with malignancies, and no karyotypic abnormalities in the family members have been previously identified. They assessed alterations in a specific area of p53 using NFS cell lines generated from members of three generations of the family. The heterogeneity of the p53 mutation recently found in noncancerous cells could mean that a family is predisposed to carcinogenesis in the early genetic changes.

Fields and colleagues[33,202–204] discussed that in the transcription activation domain, the wild-type p53 (wild-type) protein is available. The hereditary nature of the p53 mutation discovered in noncancerous cells could point to an early genetic change that predisposes a family to cancer. It has been discovered that the protein expressed by the wild-type p53 proto-oncogene inhibits transformation, whereas mutations in p53 make it transformation-competent. They discovered that p53's ability to prevent transformation is directly related to its transcriptional activity.

Cross and colleagues[88,92,93,99,205–207] describe the temperature-sensitive behavior of a specific mutant, p53val135, and p53 can regulate the cell cycle's Gl-S and G2-M checkpoints. DNA damage is responsible for complex physiology, including cell cycle disturbance and planned cell mortality, in developing mammalian cells. For cycle advancement, cell cycle checkpoints are

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required for the integration of external and internal monitoring signals. In the absence of induced DNA damage and response to environmental stress, defects in checkpoint control genes are linked to increased rates of mutation and chromosomal instability. Increased levels of expression of p53, p21CIP1/WAF1, and GADD45, as well as decreased activity of members of the CDK family, have been linked to the G1 delay in response to ionizing radiation-induced DNA damage. G2 delay following DNA damage in several immortal rodents and human tumor cells has been examined and was connected with the CDK inhibition and an accumulation of p34cdc2 proteins as tyrosine-phosphorylated versions.

Miyashita and colleagues[13,77,83,84,135,179,203,208–212] discussed AT gene(s), p53, and GADD45 are three components in the cell cycling arrest signal transduction network after damage to the genes; tumor build-up is believed to result from anomalies in that route because the protein is transcriptional transactivate genes. Mammalian cells have long been found to exhibit cell cycle delays brought on by DNA damage, but little is known about the molecular mechanisms that control these delays. Gamma radiation revealed G1 and G2 temporal arrest in cells with wild-type p53 genes, but only G2 arrest in cells with missing or mutant p53 genes, indicating that wild-type p53 played a significant role in G1 arrest through the ability of the p53 protein to transcriptionally transactivate genes like GADD45 and Mdm2.[135,179] When overexpressed, the MDM-2 gene, which encodes a putative transcription factor, increases the tumourigenic potential of cells. A p53 reactive domain of the creatine kinase muscle gene was employed to see if p53 transactivation may have an impact on the MDM-2. If the wild-type p53-expressing plasmid is cotransfected to cells that do not have an indigenous p53, its expression has been boosted. The product of MDM-2 Oncogene inhibits p53 transactivation and develops an intimate interaction with p53 protein.[179,208] The results reveal a possible positive goal for direct p53 regulation for the mdm2 gene. In turn, MDM-2 is binding to p53, which reduces the capability of the MDM-2-gene-responsive element to function as a positive transcription factor. The protein complex p53-mdm2 was found to be p53 protein in a cell line that had a temperature-sensitive variant. El-Deiry et al.[83] discovered a gene, WAF7, that encodes the largest p53 inducible transcript in mammalian cells during a screen for cDNAs induced by p53. WAf7 is the same as CIP1. In cotransfection tests, the T antigen's capacity to overcome ClP1 function may be achieved in part by interfering with p53 activation of endogenous ClP1 production. T antigen may impact ClP1 function post-transcriptionally because the additional ClP1 is expressed from a CMV promoter, a hypothesis that has yet to be investigated. Cyclin G could play many roles in the p53-mediated pathway. For starters, cyclin G may be involved in cell cycle arrest. The increased expression of p53 results in the arrest of cells in numerous cell lines, as documented.[77] The fact that cyclin G expression is induced in MEF without p21 stimulation lends credence to this theory. In this context, it's tempting to believe that cyclin G acts as a CDK inhibitory subunit (s). High-level cyclin G expression may suppress one (or more) of the GI cyclins by competing for its (their) catalytic CDK partners, acting as an anti-cyclin and giving cells a second way to regulate cyclin-CDK kinase activity in addition to p21. Finally, they conclude that this discovery will aid in the understanding of one of p53's pathways, whose inactivation is linked to the formation of human malignancies.

Miyashita and colleagues[14,72,213–223] show that Wild p53 can contribute to growth suppression either directly or indirectly by decreasing the expression of genes necessary for cell proliferation. It is thought that wild-type (wt) p53 functions as an antitumour agent because many cancer types result in related p53 gene deletions and mutations. The results indicate that overexpression of wild-type p53 reduces the transcriptional activity of numerous promoters, suggesting that the p53 protein is capable of suppressing gene transcripts (IL6, Rb, MDRI, p53, c-fos, bcl-2, thrombospondin-1, Tat, and NOS-2). However, interleukin 6 (IL-6) has been implicated as a mechanism in the improved proliferation of neoplastic cell components in many B and T cell diseases, as well as various carcinomas and sarcomas; many of these neoplasms are paired with a mutant p53 gene.[72] Recognize the trait of transcriptional repression of p53 and suggest that p53 and RB may cooperate as transcriptional repressors to limit the expression of the IL-6 gene during cellular differentiation and oncogenesis. Different types of malignancies frequently modify the "tumor suppressor" proteins p53 and the RB product of the retinoblastoma susceptibility gene. We investigated the ability of wild-type (wt) p53 and RB to suppress the IL-6 gene promoter (IL-6 promoter) and the potential therapeutic benefit of transformational mutations in these proteins. They demonstrate that cell cycle inhibition is correlated with the synthesis of human wt-p53 (form II) protein and that PCNA mRNA and protein expression are specifically downregulated in response to cell cycle inhibition.[214] Cross-resistance to hydrophobic natural products cytotoxic drugs (MDR1) gene has been associated with treatment resistance in cancer patients.[217] Oncogenes c-Ha-Ras-1 and p53, both of which are linked to the development of tumors, have an impact on the human MDR1 gene promoter. Ras and p53 mutations may activate the MDR1 gene during the

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development of tumors. Results suggest that p53 and Ras could be targets of the MDR1 gene promoter. Using an intermediate cotransfection test, it is demonstrated that p53, but not a transforming p53 mutant, inhibits the c-fos promoter's dose-dependent activity.[218] The ability of p53 to interfere with components of its basal transcriptional machinery, either directly or indirectly, may mediate its effects on the c-fos promoter, according to the findings. We looked at the p53 promoter itself to find targets for p53 transactivation. Northern (RNA) analysis and transient transfection experiments revealed that p53 regulates its transcription.[219] In the murine leukemia cell M1, the author demonstrates that a temperature-sensitive p53 promotes temperature-dependent reductions in the expression of the apoptosis-suppressing gene bcl-2 while promoting temperature-dependent increases in the expression of bax, which encodes a potent inhibitor of the Bcl-2 protein.[14] Patients with deletion of the wild-life allele of the p53 tumor suppressor gene and decreased expression of the powerful angiogenesis inhibitor thrombospondin-1 (TSP-1) were reported to have this transition in grown Li-Fraumeni fibroblasts.[221] By activating growth-regulating genes such p21WAF1/Cipl, GADD45, and cyclin G, p53 can cause G1 growth arrest, presumably to facilitate DNA repair. A novel technique by which p53 defends against DNA damage from exogenous and endogenous mutagens is a negative feedback loop in which NO-induced DNA damage results in WT p53 accumulation and subsequent repression of NOS2 expression.[223] This mechanism differs from cell cycle control, direct interaction with DNA repair proteins, or apoptosis.

Because p53 is a transcriptional activator that promotes gene induction after DNA damage, Wagner and colleagues[109,224–226] propose that p53 is a genetic switch that activates apoptosis-mediator genes. They use GHFT1 cells, which are immortalized somatotropic progenitors that express the SV40 T-antigen, which expresses a temperature-sensitive p53 mutant, to study the function of p53 in DNA damage-induced apoptosis. To trigger p53-dependent apoptosis, a transcription transactivator-independent approach can be applied.[226] Under some physiological conditions, dysregulated c-myc proto-oncogene expression might cause apoptosis. They demonstrate that the production of wild-type p53 is required for vulnerability to Myc-mediated apoptosis in primary embryonic fibroblasts that express a temperature-sensitive p53 allele ectopically but lack endogenous p53 expression. P53 may play a different role in apoptosis than it does in cell cycle arrest.[109] The ability of p53 to control transcription is assumed to play a role in its function. The ability of a protein to trigger G1 arrest is determined by the presence of transcriptional activation domains at its amino terminus. The p53 tumor suppressor protein is a transcriptional activator that can trigger apoptosis in a wide range of cell types. The apoptotic effects of several p53 deletion mutants were studied in transient transfection of HeLa cells experiment to see if sequence-specific transactivation is essential for p53-induced apoptosis. The genetic anomalies of the tumor may change the percentage of transactivation-independent apoptosis that aids in p53-mediated tumor suppression.[225]

Apoptosis driven by p53 can regulate carcinogenesis, according to Howes.[23,73,78] To study the role of p53 loss in tumor progression, researchers used an SV40 T antigen fragment that impairs pRB family activity but not p53 function to produce irregular proliferation in the brain choroid plexus epithelium of transgenic mice. In each instance, a decrease in apoptosis is related to aggressive tumor development in the absence of p53 functioning. These results demonstrate the critical role of p53-dependent apoptosis as an in vivo cancer regulator. This process happens in response to oncogenic events. In each case, a decrease in apoptosis is associated with a more aggressive tumor growth when p53 function is absent. These findings demonstrate that p53-dependent apoptosis, which occurs in response to oncogenic events, is an important cancer regulator in vivo. We adopted multiple transgenic mice expressing a weakly oncogenic fragment of the SV40 T antigen and mice exhibiting a germline disruption of p53 to study the impact of p53 loss during tumor progression. The results show that p53 inactivation's critical rate-limiting contribution to tumor growth is a decrease in apoptosis.[23] The promoter of the interstitial retinol-binding protein (IRBP) gene was used to guide SV40 T antigen production to retinal photoreceptor cells, resulting in a transgenic mouse model for retinoblastoma. This gene is activated before photoreceptor terminal differentiation. They looked into the role of p53 in the development of retinoblastoma in mice because inactivation of the retinoblastoma (pRb) and p53 tumor suppressor proteins is linked to T antigen-transforming activity.[73] In this context, p53 is required for apoptosis induction because mice expressing E7 in a p53 nullizygous background develop retinal tumors rather than degeneration. They created animals in which the IRBP promoter drives the expression of the human papillomavirus type 16 (HPV-16) E7 protein to investigate the role of p53 inactivation in retinoblastoma formation in the transgenic mouse system.[73] Tumor suppressor proteins are hypothesized to govern cell cycle regulation and play a function in mammalian development. The cellular tumor suppressor genes p53 and Rb are known to be inactivated by the human papillomavirus type 16 oncoproteins E6 and E7, respectively,

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altering cell growth control, at least in part. We created transgenic mice with E6 or E7 expression directed to the developing lens to investigate the role of p53 and Rb in murine lens morphogenesis. Microphthalmia and cataracts were seen in transgenic mice expressing E7, whereas cataracts were present in transgenic mice expressing E6.[78] These findings point to the importance of Rb and p53 tumor suppressor gene products in managing exit from the normal cell division cycle in growing lens fiber cells. Their findings also show that while neither E6 nor E7 are sufficient to enhance effective lens carcinogenesis in vivo when these two oncoproteins are combined, the lens has a greater capacity for tumor development, which is consistent with earlier tissue culture findings.

Inactivation of p53, according to Bouffler and colleagues[227,228] can result in an increased frequency of chromosomal rearrangements and mutations in p53 deficient mice. Following radiation exposure, chromosome segregation problems and a delay in mitosis in p53-deficient mice lead to a G2-M checkpoint function for p53, which is consistent with cancer data in these animals. p53 plays a role in G, cell cycle arrest caused by DNA damage and other cellular stressors in cellular in vitro systems; functional loss of p53 reduces or eliminates G, arrest after a variety of treatments, including ionizing radiation. Ionizing radiation is known to cause cancer and chromosomal aberrations; additionally, different chromosomal aberrations have been linked to specific cancers. Despite the absence of a significant overall chromosomal radiation effect in p53 defective animals, certain types of induced aberration were more common.[227] Inactivation of p53 by HPV 16 E6 after mild UV exposure leads to reduced survival and higher mutagenesis. Researchers evaluated UV-induced and spontaneous mutagenesis in clonal cell lines expressing either high-risk (HPV16) E6 or E7 or low-risk (HPV11) to investigate the mechanisms driving genomic instability in cancer. E6 HPV proteins are given to parental RKO cells, a colon cancer cell line that exclusively expresses regular p53. Using either a control vector or the HPV16 E8 (RC7.6 and RC7.14) genes, UV-induced and spontaneous HPRT (hypoxanthine-guanine phosphoribosyl transferase) mutagenesis were examined in RKO cells transfected with the HPV162 E6 (RC10.2), HPV11 E6 (RC11.6), or HPV16 E8 (RC7.6 and RC7.14) genes (RCneo). Our findings show that p53 inactivation causes a slight increase in the frequency of UV mutations in moderate UV exposures, which is consistent with p53's role in DNA repair.[228]

In 1997, Kubbutat et al.[229] discovered that MDM2 regulates p53 stability. The protein stabilizes in response to an activating signal, such as DNA damage, which results in a sharp increase in p53 levels and the inhibition of cell development. MDM2 appears to be important in controlling the pace at which the p53 protein is degraded. MDM2 inhibits the cell phase arrest of p53 and apoptotic effects and shows that contact with MDM2, due to enhanced proteasome-dependent degradation, is likely to result in a considerable decline in protein levels of p53. Since MDM2 is regulated by p53 transcription, this degrading process can help maintain low amounts of p53 in normal cells. In addition, processes affecting the degree and duration of the reaction of p53 can contribute to the regulation of p53 degradation due to MDM2. As the findings show, the fact that the mice with a functional play 53 genes have reduced gene deletion of the MDM2 genes at early embryogenesis plays an important role in controlling the p53 activity. Mice that had both MDM2 and p53 removed simultaneously showed normal development, proving that MDM2 is necessary for the negative control of p53's growth-inhibitory actions during development. Last but not least, research indicates that MDM2 sequences other than those necessary for p53 binding may contribute to the targeting of proteins for degeneration. This suggests that other MDM2-associated proteins, such as RB and E2F, may become unstable as a result of their integration with MDM2 or may change the stability of p53 through their interaction with MDM2.

Tamar published an article in 1999[230] explaining that the MDM2 gene was the target of direct transcriptional p53 activation. Several investigations have looked into the association between MDM2 deregulation and patient prognosis. The MDM2 relevant insight with p53 to create a tight unique complex, which inhibits p53-mediated transcriptional activity. Multiple molecular pathways are involved in MDM2's inactivation of p53. Since the MDM2-binding domain of the p53 overlaps the TAD, physical contact is prohibited between the p53 TAD and key transcription-binding proteins, resulting in a transcriptionally inactive p53 in the MDM2-bound form. Not just by reducing its transcriptional activity and eliminating it in the cell, MDM2 can suppress p53 function. MDM2 is crucially important for development through the negative regulations on p53 activities; p53 is probably abnormally active without the function MDM2, resulting in a fatal embryonic phenotype. Many characteristics of p53 action, including phosphorylation and acetylation, are considered to be regulated by posttranslational modifications (PTMs). Mdm2 contains many suspicious phosphorylation sites for a range of protein kinases. In vitro, casein kinase2 (CK2) and DNA-PK were also found to be phosphorylated in Mdm2. The interaction between p53 and MDM2 was probably the sole aspect of the

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intricate p53 image, which until recently was believed to be straightforward and well known.

In research published by Thomas et al.[231] described the biochemical and biological difference between the two polymorphic forms of wild-type 53 in 1999. No study comparing the biochemical and biological activities of the structurally varied wild-type variations p53 was undertaken, despite extensive research on the differences between wild and mutant p53. For this problem, the two largest variants of wildlife p53 were detected in general populations and carried out in this investigation. They observed several major alterations between them. P53Pro is discovered to be a larger transcript inducer compared to p53Arg and at least in part because the transcription factors TAFIII32 and TAFII70 are more related. There is limited knowledge about the possible differences between the two common variations in the polymorphism of wild-type p53. The first study to evaluate these two p53 variants indicated that the interactions with the simian virus of 40 T-antigen were functionally identical. Antibody reactivity-based comparisons of the p53Pro and p53Arg conformations confirm these observations. PAB1620, which was conformation-dependent, reacted equally strongly with the two versions of p53, but p53, which identifies p53, did not demonstrate reactivity.

Marin et al.[232] state that a residue of polymorphism inside p53 is impacting the mutant division, and family members' inactivation can contribute to the phenotypical characteristics of the p53 mutant's subgroup. The p53 tumor suppressor protein, which binds to certain DNA regions and begins transcription, can inhibit the proliferation of the cells and keep the genome stable. Two newly discovered mammalian p53 homologs are p73 (Encoded with TP73) and p63 (Ket, p40, p73L, p51; TP63 encoded). Both proteins bind in vitro to DNA-binding sites of canon p53 and may in vivo stimulate transcriptionally the target p53 genes at least if it is overproduced. Despite this, TP73 and TP63 are infrequently modified in human cancer and are uncertain about their specific role in human carcinogenesis. According to this study, some structural p53 mutations may bind and disable the p73 protein. Before discovering that an excess of particular p53 mutants affects the behavior of cancer cells that are deficient with p53, p73 might be explained by inactivation. They demonstrate that a common TP53 polymorphism affecting TP53 codon 72 influences the interaction of p53 and p73. The results reveal the capacity of the confirmative p53 mutants to produce stable p73 connections is affected by the common TP53 polymorphism. The loss of DNA-binding capacity of p73, and therefore its ability to function as an inductor of death and a sequence-specific transcriptor, is linked to its creation.

Following the discovery of MDM2 gene amplifications in soft tissue cancers and osteosarcomas, many researchers looked for MDM2 gene amplification and MDM2 protein overexpression in other tumors and malignancies. To present, the overall frequency of gene amplification has been estimated to be 7%, with soft tissue tumors having the highest prevalence of amplification. Amplification of the MDM2 gene has been observed in a variety of human cancers, that is, many subtypes of sarcoma, including osteosarcoma, soft tissue tumor fibrosarcoma, and so on. Momand et al.[233] gathered and outlined an MDM2 gene amplification database which is available on the WorldWide Web in 2000. MDM2 binds to the p53 protein in regular cells and maintains it at a low level, making the proteolytic proteasome more vulnerable to 26S.[233] This paper investigated the significant breakthroughs in MDM2 research during the last 12 years, as well as the possible applications of this knowledge in the battle against cancer. The first indication that MDM2 is an oncogene was the notable tumourigenic potential of an amplified genomic clone of MDM2 in rodent cells in the naked mouse. Researchers have known since 1992 that MDM2 controls the function of p53 and the frequency of MDM2 mutations in individuals.

Dumont et al.[234] demonstrated that codon 72 variations of the p53 polymorphism had functional differences in inducible cell lines with genes matched to each other and in melanoma cell lines with endogenous p53 in this study. We show that the Arg72 variant is more successful in triggering apoptosis than the Pro72 variant and that at least one mechanism for this increased effectiveness is more localized in mitochondria for the Arg72 form. In p53-dependent apoptosis, the mitochondrial location of p53 is well-known as p53 mutants that have reduced mitochondrial location are also less effective for triggering apoptosis. They found that the higher mitochondrial localization of the Arg72 polymorphic variant is caused by increased MDM2 binding and ubiquitination of the Arg72 protein, which results in increased nuclear export of the Arg72 variant. This was discovered using immunofluorescence staining of an inducible cell line. Finally, the findings imply that p53 codon 72 polymorphisms are associated with apoptotic response variability between individuals.

Cell proliferation and oncogenes, the p53 (p53, p63, and p73) and MDM2 (MDM2 and MDMX) families act as important signals integrators of genotoxic and oncogenic stress, which is discussed by Michael and Oren[235] in the year 2003. The study focussed on the intricate relationship between MDM2, MDMX, and family members p53, with a special focus on cancer concerns. The p53 function of its main negative regulator, MDM2, and of

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the MDM2 protein MdmX are affected by high activity. The MDM2 and MDMX proteins maintain yet another level of regulation on the p53 family. MDM2's ability to ubiquitinate p53 plays a critical role in maintaining P53 at very low levels in unstressed settings for proteasomal destruction. The MDM2 gene is a p53 positive transcriptional target that often has increased expression after p53 activation. This generates a negative feedback circuit in which p53 updates MDM2, compared to p53 in MDM2. Data from p53, MDM2, and their related proteins have been collected enormously in recent years and represent a real challenge for grasping knowledge.

A 2003 study about MDM2's structure, biological functions, and interactions with its binding partners was published by Iwakuma et al.[236] The double minute 2 (MDM2) mouse gene codes for a p53 tumor suppression regulator. Apoptotic and cell cycle arrest caused by p53, such as its loss or mutation, which are frequent occurrences in many types of tumors, would be inhibited by elevated MDM2 levels. The mdm2 1 gene was initially found as one of the three genes (MDM1, 2, and 3) overexpressed in a spontaneously transformed mouse BALB/c cell line more than 50 times by amplification (3T3-DM). Various tests are employed in various experiments to determine the MDM2 function, and some of the associated proteins may only regulate certain MDM2 capabilities. In short, MDM2 is a key element in cell proliferation and apoptosis regulation. A more comprehensive study of MDM2 regulators and the connection between MDM2 and its binding partners is necessary, in particular those affecting the cell cycle. A thorough MDM2-p53 pathway study and p53-dependent MDM2 functions will provide a clearer overview of cellular homeostasis mechanisms and carcinogenesis.

In the same year, Bond et al.[237] reported similar findings on the existence of the MDM2 gene as one of the most well-studied p53 target genes. According to this study, a single-nucleotide polymorphism (SNP309) in the MDM2 promoter increases the affinity of the transcriptional activator Sp1, resulting in increased MDM2 RNA and protein and, as a result, attenuation of the p53 pathway. In both hereditary and spontaneous cancers in humans, SNP309 has been associated with increased tumor formation. According to the hypothesis described here, increased MDM2 levels caused by SNP309 result in faster tumor growth in humans.

The TP53 gene is well-known as a cancer mutational target, and Nelson et al.[238] used a consecutive case series of non-small cell lung cancer to investigate common mutations in the TP53 gene as possibly contributing to cancer risk. It's been suggested that the codon 72 polymorphism influences the phenotype of TP53 mutations. It was observed that TP53 mutations predominantly occurred on the arginine allele, and the TP53 mutations are modified with codon 72, a polymorphism. The researchers used a series of cases involving patients with non-small cell lung cancer to see if TP53 mutations in the arginine or proline allele are more common and if the mutation and allelism combination result in clinically distinct phenotypes. TP53 mutations were found in 26% of the population (76/293). Mainly arginine allele mutations occurred (51/60, 85%), and a correspondingly large selection of proline allele losses (87% of LOH losses were proline loss). Furthermore, the combination of a mutation and a proline in codon 72 predicts a poor prognosis, particularly in patients with a disorder that affects the exterior of the lungs.

Bougeard and colleagues[76,239] reported on the effects of MDM2 SNP309 and Arg72Pro polymorphism on the age of tumors in Li-Fraumeni syndrome. Li-Fraumeni syndrome is one of the most fatal cancer hereditary predispositions induced by p53 gene mutations (TP53). The enhanced tumor formation of MDM2 SNP309 (TRG variation) in p53 mutation carriers was revealed to be linked. They aggregated existing genotype data for breast, colorectal, and lung cancer to produce a more robust assessment of the effect of this mutation on cancer risks. We combined data from 11 trials with 5737 cases and 6703 checks for breast cancer. We merged data from five studies for CRC with 1620 cases and 886 controls. We conducted a meta-analysis of fixed effects of seven trials for lung cancer, involving 4276 cases and 5318 controls.[239] Consequently, they assessed the effects of these two polymorphisms in 61 French carriers with the p53 germline mutation.[76] The average age of the tumor in the SNP309 G allele carrier (19.6 years) was much different in patients who are homozygous to the T allele (29 years). The median age of tumor formation in Arg-A allele carriers (21.8 years) was similar to Pro/Pro (34.4 years), while P53 contained codon 72 polymorphism. The average tumor age at SNP309 G allele carrier (19.6) was markedly different in patients who were homozygous for the T allele (2.9 years, $p < 0.05$). The mean age of tumor development was similar in Arg-allele carriers (21.8 years), with P53 codon polymorphism, to the average age of Pro/Pro (34.4 years, $p < 0.05$). Our data thus validate the effects on the age of germline p53 mutant carrier tumor of the MDM2 SNP309 G allele and suggest that the p53 72Arg allele may amplify this effect and SNP309 alone has little or no effect on the risk of common malignancies, however, it could change the time and prognosis of tumor starts.

The most frequent type of cancer in humans is nonmelanoma (NMSC). Bendesky and colleagues[15,240] published papers in 2007 and 2011 respectively and revealed a

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link between p53 codon 72 polymorphism, DNA damage, and clinical data in the Mexican population[241] (which included 204 BCC cases, 42 SCC cases, and 238 controls). P53, codon 72, is a reasonably frequent polymorphism, either a proline coding (P72) (R72). The biological features of both alleles vary: P72 is a greater p21 inducer, but R72 produces 5–10 more apoptosis. Their analysis examined whether genomic stability and UVB sensitivity in the p53 codon 72 genotypes are influenced. Finally, they discovered no correlation between the p53 genotype and baseline levels of DNA damage, oxidative DNA sensitivity, or DNA repair abilities. In vitro susceptibility to UVB-induced apoptosis was primarily increased by R72. No associations were substantially related between either p53 genotype or basal cell carcinoma (BCC), SCCs, or the two.

In the 2009 study conducted on the peripheral nerve tumor malignant MPNST by Brekke et al.[242] to identify new, clinically-effecting prognostic biomarkers (malignant peripheral nerve sheath tumor). Sixty-four MPNST patients with Type 1 Neurofibromatosis Sporades 36 and 28 had 14 cell cycle regulating proteins analyzed using tissue microarrays (TMAs) (NF1). There were obvious links between nuclear p53 expression and poor survival in NF1-associated patients. They also observed major correlations with the other proteins that were studied, although none of them was as strong as p53. For patients who have a non-return tumor size larger than 8 cm and 60 times the likelihood of dying with a positive expression in p53 during the first 5 years compared to the other patients, p53 has proved to be an independent survival predictor for the whole series of MPNST. The in situ protein expression in MPNST has received the most thorough examination to date, and it has been determined that p53 is a reliable alternative marker for treatment. The most comprehensive sporadic MPNST in situ protein profiling was presented here, so that expression levels of a range of proteins may be determined, which affect the control of cell cycles in this type of cancer.

The manually and semi-quantitatively scored immunohistochemical stains with microarray slices of formalin-fixed tissue for protein expression. Methods for digital image analysis reduce certain manual scoring limitations, however other methods like standardization are needed to provide an entire utility. Idikio[243] proposed a worldwide approach to normalizing the microarray of human normal and malignant tissue p53 expression in quantitative analysis. TMAs are vital elements for the creation and validation of clinically significant indicators of cancer. This study has employed a quantitative proteomic world normalization approach to assess its applicability in mixed normal human and cancer TMA investigation of protein expression in p53. To calculate the relationships for each core stain intensity, global normalization uses averages or medians of beta-actin. The log then converted the ratios, determined the average or median value, and eliminated the log value. In case of a lack of universal standardization of p53 protein expression, the positive tissue nuclei were 44% (42 out of 95) with median strength values and 40% (38 out of 95), with mean strengths as cut-offs. To end, the global normalization method helped designate the TMA set with a positive p53 stained. The approach used contributed to the definition of clear cuts and confirmed all negative tissue nuclei. Such methods of standardization should better define useful clinical biomarkers.

Noa et al.[241] highlight the numerous research studies in various phases of cancer inactivation of p53 and underline the importance of p53 mutations at each stage of advancement of cancer. The inactivation of the tumor suppressor p53 is linked to carcinogenesis. The p53 gene is typically changed, producing a stable mutant protein that is known to be present in cancer cells. Mutant p53 proteins not only lose their ability to inhibit tumors but also acquire oncogenic features that enable cell proliferation and survival. Interestingly, p53 mutations have taken place at distinct times in the course of transformation and have contributed significantly to the induction, promotion, aggression, and metastatic therapy of tumors. The mutations associated with TP53 are quite varied in their positions in the coding p53 sequence and their impact on the thermodynamic stability of the p53 protein. Neither the sequence specifically binding DNA nor the typical p53 goal genes are activated, however, by a high majority of the alterations. The lack of function in TP53 by a mutation can actively support the development of cancers by multiple additional mechanisms by numerous p53 mutations. There are endless ways of selecting a p53 mutation to achieve a successful cancer development that is impacted by numerous aspects like oncogenic stress, certain carcinogenic substances, LOH, DN, GOF advantages, and many more. In addition to the chronological effect on the growth and aggression of the tumor, the results of the evaluation further support the notion of the accumulation of genetic abnormalities.

Iwasa et al.[244] mentioned that the RASSF6 tumor suppressor protein regulates apoptosis and cell cycling through the proteins MDM2 and p53. C-terminal RASSF proteins can commonly interact with DNA damage through the MDM2 and p53 stabilizing processes. In this study, RASSf6 was demonstrated to reduce the apoptosis caused by UV radiation induction, VP-16, and p53 knockdown. They also showed that RASSF6 is participating in the G1/S arrest mediated by p53. RASSF6 interacts mechanically with MDM2 and enhances its autonomy. RASSF6 is, in contrast, subject to MDM2

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degradation. UV exposure lowers MDM2 expression and enhances p53 expression in the presence of RASSF6. These findings reveal a method for determining whether RASSF6 or MDM2 is more quickly degraded, as well as if DNA damage alters the balance of RASSF6 and MDM2. Even when exposed to UV radiation and p53 expression is not enhanced, MDM2 express resistance is maintained in RASSF6 deteriorated cells. All of these findings support the idea that RASSF6 acts as a tumor suppressor via interacting with MDM2 and p53.

Shi-Wei et al.[25] discovered that compound C not only induced autophagy in all cell cancer lines but also increased apoptosis in p53-positive skin cancer cells compared to p53-negative skin cancer cells. An AMP-activated protein kinase (AMPK) inhibitor has been found to induce apoptosis in myeloma, brain cancer, and glioma cells. Compound C can upregulate the expression of p53-target genes in cutaneous BCC cells that express the p53 protein and cause the overexpression, phosphorylation, and nuclear translocation of the p53 protein. They discovered that p53-knocKDown cancer cells were significantly less sensitive to compound C, with G2/M cycle arrest and lower C-induced apoptotic chemicals but no autophagy. They compared stable p53-knocKDown BCC sublines to wild-life p53-experiment BCC cells. According to their findings, compound C-induced apoptosis in the skin is dependent on the cell's p53 status. Finally, their findings show that C in the lines of skin cancer cells can cause autophagia and apoptosis.

Emmanuel et al.[245] provide an experimental comparison of the artificial neural network (ANN) assembly and nonensemble forms for the prediction of pulmonary cancer and support vector machines (SVMs). These machine learning classifiers have been taughen to prevent lung cancer by using patient nuclear samples from the epidermal growth factor receptor IGDB biomarkers, oncogenes from Kirsten rat virus, and NSCLC p53 tumor suppressor Corpus. For the training of the preferred classification instrument Voss DNA encoding has been used to map nucleotide sequences of both mutant and normal genomes. To draw from the encoded nucleotide sequence of oriented gradient (HOG) histograms and local binary pattern representative genomic features (LBP). ANN ensemble and HOG are best suited to 95.90% accuracy and a mean 0.0159 square error of the training data package of this study. The ANN ensemble and HOG genomics result in an automated test and early diagnosis of lung cancer. The outcome is very encouraging. This helps pathologists to give targeted molecular therapies and to advise patients and those in vulnerable demographics with early-stage lung cancer. They suggest an artificial network set of orientated genome gradient histograms for the prediction of lung cancer. The presented approach has seven advantages: automatic neural network prevention, several lung cancer biomarkers on one single platform, GNG compliance, and high precision prediction.

According to Arkadiusz et al.,[246] the effects of tissue decalcification on the quantification of breast cancer biomarkers using Digital image capture and processing technology innovations greatly enhance tissue protein expression measures. In particular, for decalcification treatments for bone metastatic illness, breast cancer biomarkers present a unique chance to do digital image analyses to evaluate and compare digital data and categorical results produced from the tissue preparation procedure. Tissues for up to 24 h were prospectively decalcified and stained with ER, PR, Ki-67, and p53 immunology (IHC). The pathology archive has found HER2-positive areas of breast cancer and included the category results for HER2 in the pathology reports. The results were also commented on. Leica and Aperio slide scanners captured digital photos. A Gaussian mixture model was used to convert digital to categorical scores and the accuracy was assessed by comparison to clinical scores. Digital image analysis significantly enhances protein expression quantification in human tissues.

Zhang et al.[247] used IP, immunofluorescence, clonogenic survival and cell proliferation, flow cytometric assays, and in vivo xenografts to examine the biological effects of ReACp5, a cell-permeable peptide inhibitor of p53 aggregation, on p53-carrying PCa mutants. The mutant p53 protein could create tumor cell aggregates which generate dominant-negative effects over the natural type p53 protein as well as tumor loss or novel oncogenic functions. p53 mutations are typically discovered in recurrent and metastatic diseases and are involved in initial prostate cancer pathogens (PCa). Therefore, in cases of advanced PCa where no other curative options are available, targeting mutant p53 may constitute a therapeutic alternative strategy. Our findings show that targeting mutant protein amyloid aggregates by ReACp53 restores p53 nuclear activity as a transcriptional factor, causes mitochondrial cell death, and reduces DNA synthesis in PCa mutant p53-carrying cells. The results provided here show a therapeutical potential to target mutant p53 in advanced PCa environments, which transforms how these tumors are handled and have a clinical consequence for aggressive PCa. In conclusion, their studies have shown that ReACp53 aims to accumulate amyloid and restore the nuclear function of the p53 mutant resulting in the suppression of the proliferation of cancer cells. Interestingly, we found that treatment ReACp53 has similarly modified CRPC cell reaction to AR targeting therapy with mutant p53.

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Yiming et al.[248] suggested MRI characteristics in lower-grade gliomas predict p53 status via a machine learning approach. They established a machine learning here p53 prediction model based on radiomic characteristics taken from traditional gliomas p53 status. Preoperative MR scans were taken retroactively in 272 patients with primary-grade II/III gliomas. Patients have been randomized randomly with a 2:1 ratio to training ($n = 180$) or validation ($n = 92$). Each patient has extracted a total of 431 radiological characteristics. The selected features and a SVM classification were then used to construct a machine-leaving model for predicting the condition of p53. The prediction performance of all the different functions and the model was calculated in the training and validation sets utilizing receptor operational feature curves. This approach comprised four first-order statistics containing maximum, median, minimum, consistent wavelet, shape, and size (Dipple) characteristics, and 10 textural or related wavelets. This method consisted of four first-order statistics. This process is comprised of four statistics of first-order (including minimum and uniformity) and (including correlation, run percentage, and sum entropy). Predictive accuracy was based on the area under the curve that is better than the individual characteristics in 89.6% of workouts and 76.3% in the validation system.

Overexpression of the murine double minute 2 (MDM2) gene, which suppresses p53, is rather prevalent in neuroblastoma. The p53-MDM2 pathway is regulated at various molecular levels in neuroblastoma, including through interactions with other proteins (MYCN, p14ARF). Because most neuroblastomas have a functional, wild-type p53, it's critical to either find a direct MDM2 inhibitor or target protein modulators that inhibit MDM2, allowing p53 to function more effectively. The use of MDM2 inhibitors is now being examined in preclinical and clinical research to target disease, according to Zafar et al.[249] High-throughput virtual and cell-based screening methods, natural and synthetic combinatorial libraries, biochemical and molecular biology methodologies, and other methods should all be used in the development of novel MDM2 inhibitors to treat neuroblastoma.

Gain-of-function (GOF) p53 protein has been found in cancer-associated fibroblasts, according to several investigations (CAFs). CAFs play an important role in the modification of the tumor microenvironment (TME). CAFs can be identified in almost every form of malignancy. Therefore, eliminating CAFs or reprogramming them to assume a normal fibroblast phenotype holds great promise for the therapy of cancer. GOF p53 protein may be packed into tiny EVs and delivered to fibroblasts, as demonstrated by Ma et al.[250] Small EVs can be used by tumor cells to change fibroblasts into cancerous phenotypes. HSP90 is a chaperone protein that binds to the GOF p53 protein and bundles it into tiny extracellular vesicles (EVs). EVs have become crucial cell-to-cell communication mechanisms. By demonstrating that the process of GOF p53 packing in small EVs is dependent on interaction with HSP90, they showed how to target small EVs from cancer cells while leaving WT p53 in normal cells unaffected. As a result, a combination of HSP90 inhibitors and immunological therapy for cancer patients with specific p53 mutations could be a viable alternative.

Given that most cancer cells have a faulty p53 signaling pathway, restoring wild-type (WT) p53 to cancer cells is a simple notion. Oncolytic adenoviruses are another type of viral gene therapy that is linked to p53. Several DNA viruses have been found to encode oncoproteins that inactivate p53. Polyomavirus SV40 large T-antigen, adenovirus E1B, and human papillomavirus are all examples of viral oncoproteins.[181] p53's main negative regulators are MDM2 and MDMX. Most normal tissues tolerate p53 reactivation by MDM2 and MDMX inhibitors well, even though these tissues express WT p53. TP53 mutations, deletions, and translocations are seen in the majority of tumor cells. The nature and extent of these changes differ by tumor type. Because of its critical function i.n tumor suppression, p53 has sparked a lot of interest in drug research, because any clinically effective treatment that targets the p53 pathway will save millions of lives. Although an adenovirus-based gene therapy that reintroduces WT p53 into cancer cells was approved for clinical usage in China, this strategy failed to acquire considerable momentum, owing to the inherent limitations of adenovirus-based gene delivery in tumors, as detailed in this article by Huang.[251] Furthermore, contemporary efforts to utilize the immunogenicity of the p53 protein for cancer immunotherapy will be discussed. It will be explored the challenges and future directions for targeting the p53 pathway.

Understanding how mutant p53 proteins influence cancer characteristics would be extremely beneficial to both science and patients suffering from mutant p53 tumors. The occurrence and recurrence of missense TP53 alleles have prompted a frenzy of study into the function of the mutant p53 protein because practically all TP53 mutations are thought to affect wild-type p53 activity. Kennedy et al.[20] investigate the present level of knowledge about TP53 allele variation in cancer, as well as new data that both support and refute the p53 GOF paradigm. In these and other investigations, researchers are profiling TP53 mutations more methodically. These and other studies may 1 day reveal how various TP53 mutations function as cancer drivers and whether or not

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tumors with various mutations are phenotypically distinct. Intriguingly, some TP53 mutations may acquire pro-oncogenic activity via a separation-of-function (SOF) process as opposed to a GOF mechanism.

A mutation in the p53 tumor suppressor gene is one of the genetic variations that occur most frequently in human and animal neoplasia. Among the most frequent genetic anomalies in cancers of humans (and other animals), allelic loss at the p53 locus on chromosome 17 and sequence-specific DNA binding domain missense point mutations are both present. Although p53 and its role in neoplasia are the subjects of a wealth of knowledge, Prives et al.[252] contend that it is more advantageous to give a concise summary of the "facts" surrounding the biology of p53 and its role in neoplasia, highlighting our current understanding of the p53 pathway's physiological role.

By serving as a cellular stress sensor, p53 is assumed to be a tumor suppressor. When p53 is activated in the presence of acute DNA damage, it promotes either G1 cell cycle arrest, which facilitates DNA repair, or apoptosis, which kills damaged cells. Even though p53 has a clear impact on cell-autonomous processes, evidence reveals that p53 activity in tumor cells changes the TME and supports tumor suppression in a noncell-autonomous manner. In unstressed cells, the E3 ubiquitin ligase MDM2 targets p53 for destruction, maintaining p53 at low levels. Various p53 effects may be interchangeable and thus redundant, as several pathways affecting various biological processes are expected to work together to decrease tumors. Understanding the p53 transcriptional network highlights how this knowledge might be used in cancer therapy by identifying possible vulnerabilities of p53-deficient tumors, and recent findings from mouse models and unbiased screens in the context of previous studies, argue for the importance of the pleiotropic effects of the p53 transcriptional network in cancer inhibition, according to authors Boutelle et al.[252]

p53 is activated by a variety of stressors and then controls an intricate antiproliferative transcriptional program that affects a wide range of biological responses. Cancer biologists are fascinated by p53, and its precise characterization has yielded important insights into gene control processes and nature's anticancer defences. While there is a large and sometimes contradictory body of research on p53, it is now abundantly obvious that cellular responses to p53 activation include a complex interaction of activation triggers, cell lineage, and cell state. Kastenhuber et al.[254] discussed that despite the numerous features of the p53 network that have been revealed, a full understanding of how and in what settings p53 exerts its various functions remains challenging. The consolidation of convergent truths obtained from extensive studies and the translation of p53 knowledge into the therapeutic application will be a challenge in the approaching era of p53 research.

Cancer is a disease with a wide range of symptoms. These disorders may lack a common characteristic, mutant gene, causative agent, or treatment. The p53 protein initiates a transcriptional program that results in cell-cycle arrest, repair, and restoration to homeostasis or cell death, depending on the cell type, source of stress, and level of that stress. The goal of p53 is to boost levels and activity in response to internal and external stresses, which can speed up the rate of mutations by reproducing damaged DNA or by taking part in other processes during cellular replication. Apoptosis, ferroptosis, necroptosis brought on by TNF or FAS ligand, and cellular senescence followed by immune system cytokine release are the five distinct types of cell death that the p53 gene controls. The inactivation of the p53 protein and the p53 pathway is the most prevalent spontaneous mutational inactivation event or suppression by bacterial or viral pathogenesis of cancer. Tp53 mutations are found in 50% of all spontaneous malignancies in humans, and the wild-type (WT) Tp53 gene has a deleterious impact on various p53-related functions in most, if not all, cancer cells. Levine[255] described six alternative ways to develop translational therapeutics for the p53 protein and pathway, as well as clinical studies including a huge number of potential pharmacological molecules and agents.

Because of its critical role in coordinating cellular responses to genotoxic stress, the p53 protein is known as the "guardian of the genome." The stabilization of p53 has long been recognized as a "hallmark" of the DNA damage response. Mdm2 inhibitors, like many other DNA damage reagents, are particularly effective at stabilizing and activating p53 action. Targeting p53 acetylation at the CTD, according to Kon et al.[256] findings, is a promising way to activate tumor suppression. Depending on the tissues and target genes, CTD regulates gene expression in a variety of ways. The current research looked authors on the abnormalities that cause p53KQ/- mice to die young and the tissue-specific functions of p53-KQ proteins. The author employed an established pancreatic ductal adenocarcinoma (PDAC) mouse model in which tumor formation was caused by tissue-specific expression of oncogenic mutant K-ras G12D in the pancreas in the presence of pdx1-cre to investigate the impact of p53-KQ protein expression on tumor formation. The research shows that in the absence of significant p53 stability, acetylation can trigger p53-mediated transcription and tumor suppression in vivo utilizing the acetylation-mimicking p53-KQ mouse model.

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In 2017, Bykov et al.[257] highlight new developments and several approaches in the pharmacological reactivation of mutant p53. The cellular environment significantly influences the activity of p53. The biological effects of a substance that activates mutant p53 may therefore be influenced by many factors, including cell type, differentiation status, and concurrent exposure to various types of cellular stress. As a cellular stress sensor, the p53 transcription factor detects stress factors such as DNA damage and oncogene activity (oncogenic stress). Cellular stress causes both MDM2 and p53 to become phosphorylated, which destroys their ability to connect. This causes p53 to accumulate and become activated. Oncogenic stress causes a DNA damage response involving p53, which acts as a major inhibitor of tumor development. Mutations in TP53, which result in p53 inactivation, allow for tumor cell death to be avoided and rapid tumor progression.

In this review study, Kasteri et al.[258] investigate the function of p53 as a regulator of the ternary complex, the eIF4F complex, and translation ribosome biogenesis. The translation of mRNAs is necessary for the control of gene expression, and as a result, for the control of cell division, differentiation, and apoptosis. The p53 protein controls transcription, eukaryotic initiation factors, and ribosome biogenesis. Information is transferred from mRNAs to proteins by ribosomes. The 50 UTR of the mRNA is where ribosomes are recruited during the initiation stage, which is a crucial regulatory phase of translation. The translation must start to control cell division, differentiation, and death. An important step in the control of translation initiation is the guanosine diphosphate (GDP)/guanosine triphosphate (GTP) exchange of eIF2, which is catalyzed by eIF2B to start a new round of translation. The nucleolus is the location of ribosome biosynthesis. P53 controls both pre-rRNA processing and rRNA transcription. Understanding p53's role in pre-RNA processing may lead to the identification of novel therapeutic targets in the ribosome population with altered rRNA methylation patterns.

The effects of the expression levels of TP53 pre-mRNA splicing isoforms on prognostic and predictive outcomes were investigated by Bischof et al.[259] The expression levels of p53 isoforms and patient survival are correlated in several cancers. Pre-mRNA splicing modifies the RNA that the human TP53 gene generates, resulting in at least 12 different protein isoforms that are generated to differing degrees in different tissues and under diverse physiological conditions. Based on the levels of mRNA expression of the various TP53 isoforms in the HGSOC (high-grade serous ovarian cancer) tumor tissue, the possible predictive and prognostic functions were investigated. The current study aimed to investigate the prognostic or predictive significance of p53 isoforms in a cohort of HGSOC patients with chosen patient responses or lack of response to chemotherapy. Potential predictive and prognostic functions were examined based on the levels of mRNA expression of each unique TP53 isoform in the HGSOC tumor tissue.

Zhang et al.[260] described how comprehension of mutant p53 activities led to the development of state-of-the-art strategies to promote mutant p53 degradation or restore p53 activity for forthcoming GBM (glioblastoma) therapies. P53 is typically downregulated in cancer and is essential for preserving cellular homeostasis. One of the most often dysregulated genes in cancer is TP53. The p53-ARF-MDM2 pathway is dysregulated in 94% of GBM cell lines and 84% of GBM patients. An E3 ubiquitin ligase called MDM2 causes the proteasome to break down p53, which adversely regulates it. Therefore, reactivating p53 activity by inhibiting the MDM2/p53 connection is a viable approach for the treatment of cancer and GBM. The question of whether the TP53 mutation represents an initiating driver mutation is brought up by the early TP53 mutation in secondary glioblastoma. Another point of dispute is the mounting evidence that suggests GOF oncogenic mut-p53 can accelerate the advancement of both cancer and GBM in addition to the loss of wt-p53 function. The extraordinarily high expression of GOF mut-p53 in human GBM is a potential therapeutic target for mut-p53 tumors that are rarely utilized.

The justification for numerous diverse techniques in the creation of p53 targeted therapeutics was described by Levine.[261] These kinds of therapies are developed, tested, and reviewed, and they will have a substantial impact on cancer treatments and perhaps cancer prevention. The most likely way for a mutant p53 protein to become active is by the formation of protein complexes with a cellular transcription factor, which could lead to altered chromatin architecture and new transcriptional programs in cancer cells. Every type of cell or tissue, including p53 mutants, may have a unique GOF protein interaction and necessitate a new set of medications. At least six alternative approaches to the treatment of cancer cells expressing wild-type or mutant p53 proteins have been studied. A mutant or wild-type p53 interacting protein, the mutant p53 protein, or a synthetic lethal target of a cell with a mutant p53 protein are potential targets for these techniques.

An important step in the diagnostic process is the study of protein expression in cancer cells. Mass cytometry is a relatively new technique that uses mass spectrometry detection along with metal isotope-bound antibodies to quantify up to 50 properties of each cell. When analyzing cancer patient samples, it is crucial to

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consider the expression of the p53 protein and its signaling pathway, according to Fagerholt et al.[262] who focused on the use of mass cytometry in the evaluation of hematological malignancies. The investigation of primary patient sample p53 using mass cytometry has enormous potential. It is considered superior to conventional techniques like flow cytometry in that it allows for a complete analysis of surface indicators and wide immunophenotyping paired with intracellular signaling proteins.

Klimovich et al.[263] investigations on the functional variety of p53 mutations revealed that partial loss-of-function (LOF) mutations have a particular translational influence on the course of carcinogenesis and cancer therapy. The TP53 gene product, sometimes referred to as p53, is a transcription factor that binds DNA in a way that is particular to a given sequence to regulate many transcriptional programs, all of which can inhibit cancer. DNA damage is the most noteworthy sort of cellular stress to which P53 responds. Hotspot mutants are the subject of the majority of investigations on mutant p53. When 30% of the samples bind to mutational hotspot residues, there is a total loss of transcriptional activity and mutant p53-driven chemotherapeutic resistance. The remaining 70% of non-hotspot mutants are mostly partial LOF mutants with persistent transcriptional activity.

A few of the molecular pathways by which mutant p53 proteins may carry out their carcinogenic effects are examined by Marei et al.[264] They also discuss prospective therapeutic strategies aimed at restoring tumor-suppressive p53 capabilities. In terms of possible therapeutic applications, it is particularly significant given that mutant p53 may lead to metastasis, the main cause of cancer-related mortality. In contrast to focusing solely on whether cancer expresses wild-type versus mutant p53 mutant, it is unknown whether the numerous mutations that arise on this protein have similar activity or responses to therapy. In addition to their interactions with various transcription factors, mutant p53 proteins bind to and regulate the activity of additional nontranscription factor proteins.

Inhibitors of signaling pathways that are abnormally regulated by oncogenic mutant p53 proteins are highlighted by Blandino et al.[265] as promising therapeutic approaches. One of the most significant characteristics of tumors harboring mutant p53 proteins is genomic instability, which is often found in the later clinical stages of malignancies when TP53 mutations are most common. The inherent biochemical and biophysical characteristics of mutant p53 as well as pathways that are abnormally activated in genetically harmed cells may all have an impact on p53's stability. Autophagy is a different approach where mutp53 is involved. It is well acknowledged that the primary route involved in controlling cancer cell growth and determining how cells react to various anticancer treatments is autophagy. The small molecules PRIMA-1 and MIRA-1 were created to deliberately target mutant forms of p53 and restore p53 transcriptional activity, which causes tumor cells to enter a state known as cell cycle arrest or undergo apoptosis. To reinstate the wt-p53 transcriptional activity, mutant p53 proteins can be converted back to their wild-type conformation by PRIMA-1 and its counterpart PRIMA-1MET (APR-246). Recent research focuses on the role of mutant p53 protein as a potential source of dietary-based cancer chemopreventive agents.

How MDM2 controls p53's oligomeric state can have a big impact on how it controls p53's cellular localization and degradation. Different oligomeric versions of p53 may provide novel methods for cancer therapy, according to Katz et al.[266] They also demonstrated that p53's regulation by MDM2 for both deterioration and cellular localization can be dramatically affected by its oligomeric status. The p53 tumor suppressor protein, which primarily occurs as a tetramer in the nucleus, is the protein that has undergone the most research as a regulator of transcription. The oligomeric status of p53 has been investigated using a variety of biophysical techniques, and the findings indicate that the refined full-length protein primarily exists as a tetramer. There is strong evidence that MDM2, which is involved in the export of p53 from the nucleus to the cytoplasm, preferentially translocates dimeric p53 over its tetrameric relative. Certain mutant p53 proteins that form dimers are significantly more effectively degraded by MDM2 when compared to tetrameric mutant p53. Some tumor-derived mutant forms of p53 are driven into dimer structures, which has the result in hyper-degrading mutant p53 and suppressing p53-mediated cancer cell migration.

Vieler et al.[267] concentrate on the main p53 isoforms, a tumor-suppressor protein whose malfunction frequently causes cancer. P53 isoforms Δ40p53, Δ133p53, and Δ160p53, particularly the N-terminally abridged ones, are seen in malignant tissues and are thought to be involved in carcinogenesis. They may also inactivate p53 by rapidly aggregating, which may be caused by ectopic overexpression. These isoforms affect p53 function differently and have distinct N- and C-termini. Some p53 mutations and isoforms modify p53 action and may cause a "dominant negative effect" or "GOF" by altering selectivity to other genes. In addition to these mechanisms occurring at the gene or transcription level, p53 inactivation may also occur as a result of p53 and its isoforms aggregating. Studies have revealed that the TP53 mutation status is not necessary to determine whether a

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tumor will progress and that isoform expression is independent of the mutation status. Our understanding will be enriched and will aid in the diagnosis and treatment of cancer if we can comprehend the fundamentals of p53 and its isoforms.

Mutant p53 is suggested as a potential therapeutic target and breast cancer biomarker by Duffy et al.[268] But as a target for breast cancer treatment, mutant p53 offers many appealing properties. In preclinical models of breast cancer, several of these mutant reactivating substances, including PRIMA-1, APR-246, PK11007, and COTI-2, have been investigated for possible anticancer action. Much research has looked into the possibility that mutant p53/protein overexpression in breast cancer could play a prognostic and/or therapeutic predictive function. The p53 gene mutational status or immunohistochemically determined protein cannot be advised for either predicting prognosis or therapy in individuals with breast cancer, according to the existing evidence.

The structural characteristics, biological processes, and mechanisms of p53-MDM2/MDMX are discussed by Liu et al.[269] along with a summary of small molecules that target p53-MDM2/MDMX. Murine double minute 2 (MDM2, also known as HDM2) and murine double minute X (MDMX, also known as HDMX or MDM4) are significant p53 inhibitors and target genes downstream of p53. MDM2 contributes to the suppression of cell growth, induction of apoptosis, and regulation of the cell cycle by attaching to the N-terminus of the p53 transcriptional activation domain. MDM2 is a controller of the downstream gene of p53, which is involved in the regulation of the stability and activity of p53 protein. While p53-MDM2, MDMX negative feedback regulation, and the mechanism are of enormous significance in the normal physiological activities of cells, tumor formation, progression, treatment, and prognosis, the tumor suppressor gene p53 plays a significant role in carcinogenesis and development. In addition to improving our knowledge of the function and mode of action of the p53-MDM2/MDMX protein, the development of highly selective p53-MDM2/MDMX small molecules offers the potential for the development of novel anticancer drugs.

The most recent research on mutp53 protein accumulation, GOF, and targeted therapy for mutp53 in human cancer is summarized in this review by Yue et al.[270] A growing body of research has shown that many tumor-associated mutant p53 proteins not only lose the tumor-suppressive activity of wild-type p53 but also acquire novel activities that promote carcinogenesis independently of wild-type p53, a process known as GOF. In general, mutant p53 is unable to recognize and attach to p53 DNA-binding sites and loses transcriptional activity toward target genes regulated by wild-type p53. If mutant p53-containing proteins are targeted, patients with tumors harboring mutant p53 may have a better prognosis and live longer. It will be easier to create efficient therapeutic approaches that target tumors carrying mutp53 if we have a better knowledge of the mechanism of mutp53 accumulation in tumors.

Wang et al.[271] offered a perspective on the design and development of MDM2 small molecule inhibitors, early clinical evidence for some MDM2 small molecule inhibitors, and the clinical development of MDM2 inhibitors for the treatment of cancer. Due to its critical role in the regulation of several cellular activities, the level and activity of p53 are closely controlled. p53 is inhibited by MDM2 in several ways that depend on interaction with the protein. The fact that MDM2 is carcinogenic when overexpressed is consistent with its function as a powerful inhibitor of the p53 tumor suppressor function. Peptides or nonpeptide small compounds can interfere with the protein-protein interaction of MDM2-p53 by raising the p53 protein and activating its transcription. RG7112 was the first MDM2 inhibitor to start clinical trials in humans. Future clinical investigations of MDM2 inhibitors should consider late hematological harm, particularly thrombocytopenia, in light of the clinical results for RG7112. introductory stage Clinical evidence for three of these drugs demonstrates that MDM2 inhibitors have acceptable safety profiles and can activate patients' wild-type p53 (RG7112, RG7388, and SAR405838). Preclinical studies have shown that MDM2 inhibitors are only effective against tumors with wild-type p53, so it is expected that tumor cells with mutant p53 will be resistant to MDM2 inhibitors.

Mutant p53 protein has been identified as a key prospective target for the creation of cutting-edge anticancer drugs, according to Duffy et al.[272] Due to its high mutation frequency and essential role in initiating the development and progression of cancer, mutant p53 is a high-priority target for anticancer therapy. PRIMA-1 and APR-246 are the chemicals that have been investigated the most. After screening a library of low-molecular-weight substances for their capacity to convert mutant p53 to its active conformation, induce apoptosis, and exhibit sequence-specific binding, PRIMA-1 was first identified. Some investigations concluded that PRIMA-1 and APR-246 can operate independently of the mutant protein, even though the majority of studies showed that these drugs specifically inhibited cancer cells expressing mutant p53. Numerous research studies have revealed the anticancer activity of APR-246, and early findings from clinical trials suggest that it is reasonably benign. In addition to reactivating mutant p53, APR-246 has been

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demonstrated to bind to and activate the p53-homologous proteins, p63 and p73.

The functional importance of the relationship between mutant p53 proteins and autophagy in the development of cancer is highlighted by Cordani et al.[273] This information is crucial for creating more potent treatments for tumors with mutated TP53 genes. During autophagy, an intracellular degradative process, damaged macromolecules, and organelles are transported to lysosomes. Mutp53 proteins impede several autophagy mediators at different phases of the procedure, including Beclin1, LC3 cleavage, AMPK signaling, and ATG12. When the body is experiencing metabolic stress, a highly evolutionarily conserved serine/threonine protein kinase complex called AMPK promotes autophagy by phosphorylating several downstream transcription factors and metabolic enzymes. AMPK and wtp53's functional relationship is a well-known tumor-suppressing mechanism. The downregulation of AMPK signaling is one of the most important metabolic alterations that occur in cancer cells with GOF mutp53s. Autophagy can be prevented by mutp53-induced AMPK suppression in several ways, such as through regulating the mTOR complex, ULK1 kinase, or Sirt1 deacetylase.

The target p53 aggregation in cancer is highlighted by Kanapathipillai,[274] who also suggests using tiny stress molecules as prospective p53-anti-aggregation medications. More evidence in favor of treating cancer as a protein aggregation disease comes from recent studies on the prevention of p53 aggregation and tumor growth utilizing polyarginine analogs and designer peptides. More frequently, p53 loses its ability to bind DNA by a single point mutation (contact mutant) or structural changes to the protein's central DNA-binding domain (conformational mutants). Small stress chemicals like polyarginine and its equivalents cause p53 aggregation suppression in both in vitro and p53 mutant cancer cells. Examples of small stress chemicals include ectoine, hydroxyectoine, and mannosylglycerate, which are known to stabilize proteins under extreme stress and help reduce peptides that lead to neurodegenerative disorders. Small stress molecules could be able to control the aggregation of p53 mutants. Because arginine, its analogs, and cholinergic chloride have anti-aggregation properties for p53 mutants, small molecule protein stabilizers may have specific inhibitory effects on the aggregation linked to p53 mutations. A great treatment strategy for treating p53 mutant aggregation-related cancer may involve creating a small molecule formulation with the aforementioned characteristics.

A crucial aspect of the cancer cell's dependence on mutp53, according to Mantovani et al.[275] is its ability to recognize stress-related cell-intrinsic and extrinsic transformation circumstances. Mutant p53 proteins may promote cancer cell survival and tumor growth by acting as homeostatic regulators that detect and guard against stress stimuli related to transformation, such as DNA lesions, oxidative and proteotoxic stress, metabolic imbalance, interactions with the TME, and immune system interactions. Mutp53 controls adaptive responses that aid in the development of tumors. The adaptive responses provided by many oncogenic and non-oncogenic processes are essential for cancer cell survival, and the stress phenotypes connected to carcinogenesis have been hypothesized to represent a common hallmark of cancer. Based on drug combinations that either directly or indirectly target mutp53-dependent homeostatic processes, we predict a wide range of therapeutic options for anticancer therapies. Tumor vulnerabilities may vary based on the specific TP53 mutation and the type of tumor, as different TP53 missense mutations have various functional effects.

There is multiple evidence that the p53 protein plays a role in both the regulation of the innate immune system and as an antigen in adaptive immune responses. Immunotherapy is used to cure cancer. The p53 field attracts immunologists who will clarify the complex interactions between the p53 pathway and the immune system, claims Levin.[276] All living things require an immune system to function. Clonal cellular replication is how the immune system multiplies its output, and p53 inhibits cellular replication when it senses stress. However, to use innate and adaptive responses, p53 requires NFK-beta. It is considered stressful when foreign DNA, RNA, and proteins emerge in or are produced by the organism. This common antagonist links the p53 response and the immunological response. As the innate immune system evolved, responding to stresses and infectious diseases, invertebrates' p53-like protein likely comprised innate immunological responses as part of its signal transduction pathway. The p53 protein in its wild-type form is a component of the innate immune response. Even when the p53 protein has a mutation and is not functioning, the transcription of repetitive DNAs within the cell triggers an innate immune response.

Monti et al.[277] will explore the effects of TP53 mutations on P53 protein function and their possible effects on clinical outcomes in 2020. The main topic of discussion will be chronic lymphocytic leukemia (CLL), which has also generated a lot of debate on how TP53 mutations impact available treatments. Significantly reduced transactivation capacity in P53 proteins has been associated with more severe cancer predisposition syndromes (e.g., LFS). Various substances have been developed during the last two decades to either eliminate

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TABLE 3 Methodology description used for p53

SR no.	Methodology used	Methodology description	References
1	IP	IP is a technique for removing a protein antigen from a solution using an antibody that binds to that protein selectively. This method can be used to isolate and concentrate a specific protein from a sample that contains thousands of them. Flocculation and precipitation processes are two procedures.	[3,4,17,90,111,113,114,116–121,125,-169,180,202]
2	RIA	An RIA is an immunoassay that uses radiolabeled substances to create immune complexes in a stepwise manner. An RIA is very sensitive in vitro assay technique for detecting substance concentrations, typically antigen concentrations (e.g., hormone levels in the blood) using antibodies.	[3,117]
3	Southern blot technique	A southern blot is a method for detecting a particular DNA sequence in a blood or tissue sample. Breaking down a DNA sample into bits that can be sorted by gel electrophoresis is done with the help of a restriction enzyme.	[129,141,142,144–146]
4	Western blot technique	The western blot (also known as the protein immunoblot) or western blotting is molecular biology and immunogenetics analytical technique for detecting specific proteins in a sample of tissue homogenate or extract.	[10,34,118,125,132,256]
5	RNA/northern blot analysis	A northern blot is a lab technique for detecting individual RNA molecules in a mixture of RNA molecules. Northern blotting is a technique for analyzing RNA from a certain tissue or cell type to determine the RNA expression of specific genes.	[122,123,141,144–146]
6	Electrophoresis	Electrophoresis is a technique for separating DNA, RNA, and protein molecules depending on their size and electrical charge.	[89,115,117,129]
7	Autoradiography	An autoradiograph is a picture created on x-ray film or nuclear emulsion by the pattern of decay emissions from a radioactive substance distribution.	[119]
8	Immunoperoxidase	Immunoperoxidase is molecular biology, medical research, and clinical diagnostics immunostain. Immunoperoxidase reactions, in particular, are a type of immunohistochemical or immunocytochemical method in which antibodies are detected through a peroxidase-catalyzed reaction.	[119]
9	Calcium phosphate co-precipitation	Calcium phosphate co-precipitation is a technique that includes combining DNA with calcium chloride in a buffered saline/phosphate solution to produce a calcium-phosphate–DNA complex.	[10]
10	Metabolic labeling	Metabolic labeling refers to approaches for incorporating detection or affinity tags into biomolecules using the endogenous synthesis and modification machinery of living organisms.	[10]

(Continues)

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TABLE 3 (Continued)

SR no.	Methodology used	Methodology description	References
11	SDS polyacrylamide gel electrophoresis	SDS-PAGE is a discontinuous electrophoretic technology invented by Ulrich K. Laemmli and widely used to separate proteins with molecular weights ranging from 5 to 250 kDa.	[128,129]
12	Quantitative and qualitative immunochemical techniques	Immunoassays can be qualitative or quantitative, depending on the assay format. Identifying antibodies and antigens in a sample is a completely qualitative approach. However, there is a quantitative approach known as rocket immunoelectrophoresis.	[126,134]
13	Phosphorylation	The addition of a phosphoryl group to a molecule is referred to as phosphorylation in chemistry. Many cellular activities in biology rely on this mechanism and its inverse, dephosphorylation.	[12,158,159]
14	Southern filter hybridization	Southern filter hybridization is a method for finding homologous or complementary sequences in DNA or RNA that involves allowing a denatured radio-labeled "probe" nucleic acid (in solution) to anneal to a denatured nucleic acid (immobilized on a nitrocellulose filter) to be evaluated.	[140]
15	Cytogenetic analysis	The study of the banded pattern of chromosomes during the metaphase of the cell cycle is known as cytogenetic analysis. Because it is done cell by cell, mosaicism can be easily seen when just a few cells have chromosomal defects.	[142,183,187–189,195]
16	Chromatography	Chromatography is a laboratory technique for separating a mixture in chemical analysis. The combination is dissolved in a fluid known as the mobile phase, which transports it through a system containing a stationary phase material.	[171]
17	RFLP analysis	Using polymorphisms in homologous DNA sequences, RFLP is a molecular biology technique that can be used to identify individuals, groups, and species as well as to pinpoint the locations of specific genes within a sequence.	[6,187–189,195]
18	Methylcholanthrene modifies	Methylcholanthrene, a highly carcinogenic polycyclic aromatic hydrocarbon, is created when organic molecules burn at high temperatures. Methylcholanthrene is known by the designations 3-methylcholanthrene, 20-methylcholanthrene, and the IUPAC nomenclature 3-methyl-1,2-dyhydrobenzo[j] aceanthrylene.	[191]
19	Gamma-irradiation	A gamma-ray, also known as gamma radiation, is a type of penetrating electromagnetic radiation created by the radioactive decay of atomic nuclei. It has the maximum photon energy because it is made up of electromagnetic waves with the shortest wavelengths. In the gamma irradiation technique, cobalt 60 radiation is used to destroy germs on a range of goods in a specially constructed cell.	[135,179]

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TABLE 3 (Continued)

SR no.	Methodology used	Methodology description	References
20	Cotransfection tests	Cotransfection is the transfection of two nucleic acid molecules at the same time, such as plasmid DNA and siRNA. Cotransfection is a typical method for achieving long-term transfection stability. The plasmid DNA may contain a testable gene that serves as a marker. A marker gene is cotransfected for this to happen, giving the cell a selective advantage, such as resistance to a specific toxin. It's critical to choose the correct cotransfection reagent.	[14,226]
21	Meta-analysis	A meta-analysis is a statistical analysis that brings together the findings of several scientific investigations. When several scientific studies are addressing the same subject, and each research report measurements that are predicted to have some degree of inaccuracy, meta-analyses can be performed.	[239]
22	TMAs	TMA is the process of moving several tissue sections, usually from histologic paraffin blocks, so that tissues from multiple samples can be analyzed on the same slide.	[240,242]
23	Clonogenic survival	A clonogenic assay is a cell biology tool for determining how successful specific drugs are at promoting cell survival and proliferation.	[247]
24	Flow cytometric analyses	Flow cytometry is a technique for quickly evaluating single cells or particles suspended in a buffered salt-based solution as they pass through one or more lasers. Each particle is evaluated for visible light scattering and one or more fluorescence properties.	[91,247]

Abbreviations: IP, immunoprecipitation; RFLP, restriction fragment length polymorphisms; RIA, radioimmunoassay; TMAs, tissue microarrays.

mutant P53 that has accumulated in cancer cells or restore the WT (wild type) function. The creation of novel anticancer therapies has been thought to be a promising target for the mutant P53 protein. P53 deficiency unquestionably has a role in the clinical onset of CLL. The choice of treatment for CLL may be influenced by the discovery of molecular biomarkers as well as specific clinical aspects of the disease.

Treatments for malignancies that contain mutant p53, according to Hu et al.[278] focus on selectively targeting mutant p53, restoring wild-type activities, and using interactions between mutant p53 and noncoding RNAs that serve as targets for mutant p53 in synthetic lethal interactions. Both the anticancer efficacy of TP53 and the carcinogenic features of the mutant p53 protein is compromised by TP53 mutations. The p53-targeted therapeutic strategy started with the discovery of drugs that could reinstate/reactivate the functions of wild-type p53 or eradicate mutant p53. The structure and drug species of treatments that specifically target mutant p53 are very important. Depending on the p53 state, p53 treatments may involve stopping mutant p53 (mutp53) from functioning normally, suppressing mutant p53 (wtp53), or reversing the effects of mutant p53. Several approaches can be used to target wt/mutp53 in cancer therapy. Clinical trials show that the MDM2/X inhibitor is beneficial for tumors that express wtp53. Numerous cancer cells with mutp53 exhibit improved mitochondrial capabilities, and mutp53 preferentially demonstrates functions that promote glycolysis. Since TP53 is a popular therapeutic target for the treatment of cancer in general, the development of synthetic lethality using mutp53 would greatly increase the clinical options available to cancer patients.

Chiang et al.[279] explore the creation of small molecules or immunotherapeutics that target p53-R175H and therapies that target p53-R175H. One of these treatments involves employing tiny compounds

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TABLE 4 Data set used for p53 (animal data set)

Sr. no.	Animal data set	References
1	1. Anti-T sera 2. Hamster anti-T sera 3. Mouse anti -T sera 4. Anti-U sera raised in 5. Monkeys, SV40 transformed mouse cell line 5. SV40 transformed hamster cell lines	[111]
2	Cells of BALB/c origin were studied 1. Methylcholanthrene (MC)-induced sarcomas 2. Leukemias 3. Virus-transformed cell lines 4. Spontaneously transformed fibroblasts 5. Nontransformed cells	[3]
3	Animal species and tissues and cells (SV40 transformed cell lines used) 1. Mouse kidney (SVKM Cl. 9,11). 2. Rat embryonic lung (v11 F1 cl.1, v15 FP1 cl.1) 3. Chinese hamster kidney (CHK-SVLP) 4. Syrian hamster (E1, TSV5 cl. 2. TSV11, EHSVi) 5. human skin (WI98 VAD) 6. African green monkey kidney (BSC-SV Cl. 1) 7. Mouse embryo (ME ch V11 cl. 1-Ls, ME ch. V11 cl. 1-A) 8. Rat embryo (RE ch. V11 cl. 5-LS, RE ch. V11 cl.6-LS)	[4]
4	Animal species and cells 1. Mouse (SV3T3, H615, VLM, IC21, METH A, Py6, Py3T3, Meth 113, MSV-3T3, L, 3T3) 2. Rat (SV40rat, SV40/PyV rat) 3. Syrian golden Hamster (SV28) 4. Human (SV80) 5. Monkeys (SVTC-7)	[112]
5	1. Mice. BALB/cN female mice between 8 and 12 weeks of age 2. Cell lines of A-MuLV-transformed cell lines of BALB/c 3. 70Z, a BALB/c chemically induced pre-B-lymphocyte line 4. NIH/3T3 derivation 5. A2, A-MuLV-transformed nonproducer 6. A2/M, A2 superinfected with M-MuLV	[113]
6	1. Cell lines-Hela cell lines of mice used 2. Mouse kidney cells 3. tumor sera (Hamster)	[116]
7	1. Rat embryo fibroblast 2. Chinses Hamster embryo fibroblast 3. Anti-p53 monoclonal antibodies RA3-2C2	[8–10]
8	1. Adult rat chondrocytes 2. Wistar adult xiphisternum chondrocytes	[10,125]
9	1. Molony helper virus 2. The Ab-MuLV-transformed cell lines	[126–130]

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Sr. no.	Animal data set	References
	3. Monoclonal antt-p53 antibodies hybridoma cell lanes RA3-2C2 4. Rat-1 cells 5. Mouse cells 6. Monoclonal anti-p53 antibodies hybridoma cell lines RA3-2C2 and PAb122 7. The MB 63/T/36 (BL) bladder carcinoma cells(mice) 8. Nude mice	
10	1. Ab-MuLV-transformed lymphoid cell lines 2. Antibodies 3. Friend leukemia virus 4. Friend murine leukemia virus	[131,132]
11	1. Monoclonal antibodies PAb248 and PAb242 2. Chinese hamster gadd45 clone 3. Friend leukemia virus 4. Friend murine Leukemia virus 5. Hematopoietic tissue cells 6. Human myeloblastic leukemia cell lines 7. Bone marrow cells 8. Chinese hamster B-actin clone	[21,131,133–135]
12	1. Cell lines (Ab-MuLV-transformed cell line) 2. Antibodies (monoclonal anti p53)	[140–142]
13	1. Monoclonal antibodies 2. Clone 6 of rat embryo fibroblast 3. Mouse mutant 4. Monkeys COS cells 5. Rabit antiserum 6. Amino acid from human protein 7. Rabbit reticulocyte lysate 8. HeLa cell 9. Anti-hTAFII31 polyclonal antibodies 10. pGST-p53 1-73	[159,162–175,177,178]
14	1. Monkey's COS cells 2. Mouse wild-type cells	[17]
15	1. SV40 T antigen 2. Wild-type purine p53	[18]
16	1. Anti-T monoclonal antibodies 2. HeLA cell 3. Anti-pol antibodies	[180]
17	1. Plasmid	[22]
18	1. Primary rat embryo fibroblasts 2. Plasmids	[189]
19	1. E6 protein 2. Wild-type E6 3. Rabbit reticulocyte lysates	[191]
20	1. Cell lines Ab-MuLV-transformed lymphoid cell lines 2. Antibodies	[193]
21	1. Mouse Friend erythroleukemia cell line 2. BALB/c mouse liver DNA	[34]

(Continues)

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TABLE 4 (Continued)

Sr. no.	Animal data set	References
22	1. Cell line in athymic nude mice	[194]
23	1. Plasmids 2. Primary rat embryo fibroblasts	[195]
24	1. Primary rat embryo fibroblasts 2. Plasmids	[192]
25	1. RKO cell lines	[6]
26	1. Plasmids	[90]
27	1. CAT gene containing the Elb TATA 2. Five types of plasmids	[33]
28	1. HeLa cells 2. CAT plasmids	[201]
29	1.C2 myoblast cell lines 2. Transgenic mice	[202]
30	1. Plasmids pLTRp53cGva135	[91]
31	1. Cultures of embryonic fibroblasts from p53	[203]
32	1. Mouse strain 2. Mouse embryonic fibroblasts	[87]
33	1. A normal mouse testis cDNA library 2. A normal mouse testis cDNA library 3. Rat embryo fibroblast	[207]
34	1. A rat embryo fibroblast 2. The MDM-2 cosmid and the vector for this murine gene CV001	[208]
35	1. Mouse cell lines 2. Monoclonal antibodies (mouse p53 assay)	[116]
36	1. Swiss 3T3 cells obtained from Andrea Mastro 2. Two p53 monoclonal antibodies of mouse origin 3. Sera of hamsters	[89,117]
37	1. Mouse thymidine kinase-deficient cells (B82) 2. Chinese hamster 3. Monoclonal antibodies 4. Several hybrid cell lines 5. Somatic cell hybrids 6. Rat spleen cells 7. Abelson murine leukemia virus 8. Mouse sarcoma cells	[139]
38	1. Monoclonal anti-p53 antibodies 2. 2M3, an Ab-MuLV-transformed lymphoid cell line 3. Monoclonal antibody PAb421 4. The Ab-MuLV-transformed cell line 5. Hybridoma cell lines RA3-2C2 6. Rat 7. Hamster 8. The L12 murine p53 nonproducer cells 9. Abelson murine leukemia virus 10. Hybridoma cell lines RA3-2C2 and PAb421 11. Epidermal cells of strain 291	[143–145]

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Sr. no.	Animal data set	References
39	1. Mouse cell 2. Cell lines and culture 3. Rar embryo fibroblast 4. Antibodies 5. HeLA cells 6. The hybridoma line PAb416 7. SV40 T antigen, 3T3 cells	[12,146–157]
40	1. Expression plasmids 2. Antibodies 3. Yeast strains	[83]
41	1. Cyclin D1-associated proteins 2. DNA tumor virus	[209]

that can bind to p53-R175H to reactivate p53-R175H. Hotspot mutations cause the encoding of mutant p53 proteins, which gain functions that promote the growth of cancer while losing the classical wild-type p53 capabilities. These include promoting metabolic reprogramming, angiogenesis, invasion, migration, proliferation, and treatment resistance in cancer cells. p53-R175H develops oncogenic capabilities through interactions with multiple proteins, despite being more prone to aggregation and lacking the transactivating capacity of the wild-type p53. The target genes are prevented from being expressed when p53-R175H links with other transcription factors like p63, p53, ETS2, and NF-Y. Many small compounds that potentially reactivate p53's tumor-suppressor capabilities were found via screening about therapies that target the p53-R175H mutation. It is now known that p53-R175H peptides are expressed on the cell surfaces of cancer cells and can be differentiated from the wild-type p53 peptide by T cells, making them a potential target for cancer immunotherapies.

At both the protein and mRNA levels, Rojas et al.[280] assessed for the first time the relative expression of p53 isoforms in multiple myeloma (MM). According to various studies, P53 isoforms are variably expressed in human malignancies and have been found to affect the prognosis of some of these tumors. Reverse transcription PCR measurements of mRNA levels provide the basis for the majority of the data currently available on the expression of p53 isoforms in tumors. More than 70% of MM samples had the p53 protein expressed, with long isoforms contributing more than short isoforms, which are only seen in a small number of samples. The inverse prognosis was linked to low and high levels of short and TAp53β/γ isoform expression, respectively, in MM patients. The present classification of cytogenetic risk was enhanced by taking expression levels of p53 isoforms into account. Patients with high cytogenetic risk had significantly higher levels of TAp53/isoform expression than did patients with low risk. There were no differences in the expression of the total p53 protein or any of the other p53 isoforms in either cytogenetic-risk group.

According to the review of Hibino et al.[281] numerous candidate compounds for anticancer therapies have already been developed as a result of considerable research into p53 as a potential target for anticancer drugs. Inhibiting the unique connection between MDM2 (MDM4) and p53 results in increased p53 activity in cancer cells. Although the medications' molecular targets differ, two drug discovery techniques have recently surfaced with the aim of "rescuing p53 protein." Several biophysical approaches to protein characterization were used to carry out this approach. Even in cancer cells without TP53 mutations, two main mechanisms—proteolytic inactivation and aggregation-related mechanisms—lead to "complete" inactivation of p53 activity. For the synthesis and identification of p53-MDM2 inhibitors, a structure-guided approach utilizing conventional X-ray crystallography and binding mode analysis of the seed compounds using cutting-edge NMR techniques was heavily utilized. As a result, since this strategy avoids p53-MDM2 binding and aggregation, it is projected to have fewer adverse effects. The best p53-targeting medications may therefore be those that stop p53-MDM2 binding as well as aggregation.

Zawacka-Pankau[282] provides comprehensive coverage of the most recent facts on the structure, function, and prospective targeting of the p53 protein family for improved cancer therapy. The p53 protein family has one member, P53, which has undergone the most in-depth characterization. Both transcription-dependent and transcription-independent effects are known to be produced by it. P53 is a member of a family of proteins that also contains the proteins p73 and p63. Through

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TABLE 5 Data set used for p53 (human Data set)

Sr. no.	Human data set	References
1	1. Sera blood samples were taken from the patients and volunteer control women 2. SV4O-transformed human fibroblast cell line SV80 and the mammary carcinoma cell line MDA-MB-157	[114]
2	1. Human cell lines 2. Monoclonal antibodies (human p53 assay)	[116]
3	1. Human peripheral blood lymphocytes	[118]
4	1. Human clones 2. Human fibroblast	[136–138]
5	1. Human CEM cells 2. Human CDC2 protein kinase	[158]
6	1. Human tissue samples	[183]
7	1. Neoplastic cells 2. Nonneoplastic cells 3. DNA from tumor and normal colonic mucosa of each patient	[184]
8	1. 100, consecutive resectable primary breast tumors	[185]
9	1. DNA probe on Chromosome 17 2. DNA from 58 carcinoma specimen	[182]
10	1. 30 lung cancer cell lines 2. Samples from normal lungs	[187]
11	1. Reagents 2. Tumor samples	[188]
12	1. Bone marrow 2. P53 transcripts	[186]
13	1. DNA probe on Chromosome 17 2. DNA from 58 carcinoma specimen	[6]
14	1. Osteosarcoma cells 2. Skin fibroblasts 3. Normal tissue adjacent to the tumor	[120]
15	1. Normal skin fibroblast cell line	[74]
16	1. NHF1 and NHF3 are human fibroblast cultures	[205]
17	1. DAH041 postcrisis cell line fibroblast	[206]
18	1. Human diploid fibroblasts	[83]
19	1. SW480-IAB3 cell line 2. PG13-Luc and MGIS-Luc plasmids	[82]
20	1. Human fibroblast	[209]

transcription-dependent or -independent activities, all members of the p53 protein family can trigger apoptosis in response to UV-induced DNA damage. Each member of the p53 protein family expresses different isoforms and has a unique domain structure. One of the most likely causes of the observed trend is that the chromatin's epigenetic state adjusts the transcriptional programs that are affected by p53 by controlling the presence of co-regulators, the allocation of p53-responsive elements (RE), and p53 PTMs.

In patients with CRC, Kim et al.[283] compare the clinical impact of TP53 status assessed by IHC and next-generation sequencing (NGS) and look at the association between p53 expression by IHC and TP53 mutational status by NGS. TP53 is among the main driving genes in human CRC. IHC is the approach used most commonly to assess the TP53 mutational status (IHC). The TP53 mutation is found in 43.28% of CRCs, making them the most common type of cancer to have. The expression of the p53 protein is thought to provide more information about the prognosis of CRC patients than the mutation itself, and CRC patients without p53 expression had a greater efficacy.

We sought to comprehend the entire functionality of protein p53 using diverse approaches and different sorts of datasets used for tests done by the authors to get extremely deep information about p53 protein through literature survey articles done from the year 1979 to the year 2022. Below Table 3 details the many methodologies used in P53 details employed and their descriptions, whereas Tables 4 and 5 cover the various types of datasets used, such as human and animal datasets respectively.

10 | METHODOLOGY USED

A technique is a systematic strategy to resolve a study topic by collecting data using various methods, evaluating the data, and drawing conclusions from the data in research. There are several different methodologies utilized in this review study, which are listed in Table 3 methodology used for p53.

11 | DATA SET USED

A chemical or information obtained from live organisms and their products is referred to as biological data. Text, sequence data, protein structure, genetic data amino acids, and linkages are only a few examples of biological

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data. Animal and human datasets were used for experimental purposes in all of the respective reference publications, as shown in below Tables 4 and 5: Data set used for p53.

12 | CONCLUSION

The molecular biology of p53 protein is complex, as this review has demonstrated. Here, from its discovery in 1979 till today, we discussed the P53 protein. Before it was regarded as a crucial tumor suppressor protein, P53 was an absolute non-negligible member of the protein study. P53 has been at the forefront of cancer research since its discovery in 1979. p53 protein has been one of the most explored molecules in the field of biomedical sciences throughout the majority of researchers' lives. While this is not surprising given p53's prominent function in a variety of stress responses, the p53 network's intricacy makes it difficult to completely comprehend its biology and use this information for diagnostic, prognostic, or therapeutic purposes. Researchers are still attempting to figure out whether the p53 tumor suppressor protein is a promising protein for studying and understanding early cancer detection.

AUTHOR CONTRIBUTIONS

The study ideas, as well as the formation or modification of overarching research goals and aims, were contributed by all authors. Preparation for specifically critical review, material preparation, data collection, and revised work was performed by Manisha Patil and Dr. Anand Bihari. All authors read and approved the final manuscript.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Not applicable.

ETHICS STATEMENT

The authors of this article give their consent for publication.

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